Orthopedics - Introduction and Upper Extremity

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Acknowledgements

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Most importantly, thank you Dr. Heller for allowing me to collaborate on this lecture!

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I am available to address any questions/comments/concerns and provide additional resources or discussion.

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Learning Objectives

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1. Recognize bony and soft tissue injuries to the upper extremities
2. Appreciate complications related to upper extremities
3. Understand management options in upper extremities injuries
4. Identify urgent and emergent upper extremities conditions
5. Interpret and apply clinical signs and symptoms of upper extremities processes in the differential diagnosis.

General Approach

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- In any orthopedic concern, anatomy is extremely important!
- Consider other issues in the surrounding neurovascular structures

Complications:
- Avascular necrosis: death of bone due to lack of blood supply
- Nerve damage/entrapment, and their clinical signs
- Compartment syndrome
- Infection, Osteomyelitis
- Deep Vein Thrombosis (DVT)

The Interview:
- **History/Mechanism**: High-impact trauma vs non-contact vs overuse
- Physical exam/special tests
- Diagnosis: Imaging (X-Ray, MRI, CT, US, etc)

Treatments: Start Conservative!
- RICE*, Pain control, joint injections

Surgical techniques are beyond the scope of these lectures, but the basic concepts involve reduction of displaced fractures using nails, pins, plates, and screws. This is sometimes done manually prior to surgery, or for dislocations

The Basics

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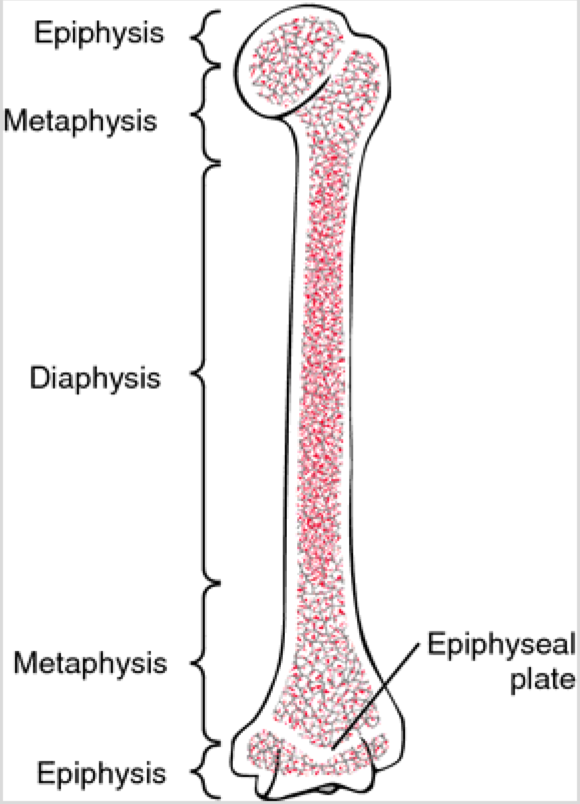
Anatomic Locations:
- **Epiphysis**: the end of the bone
- **Metaphysis:** flared portion of bone between diaphysis and epiphysis (growth plate)
- **Diaphysis**: the long shaft of the bone

Fracture Terminology:
- **Transverse**: fracture is perpendicular to the long axis of the bone
- **Oblique**: fracture down the long axis of the bone
- **Spiral**: a complex fracture line that encircles the shaft of the bone (twisting)
- **Comminuted**: more than 2 bone fragments
- **Avulsion**: fragment of bone torn away from the main mass of bone
- **Impacted**: bone fragments driven into each other, causing bones to wedge together

- **Nondisplaced**: fragments are aligned
- **Displaced**: fragments are separated
- **Compound**: displaced fracture that pierces the skin



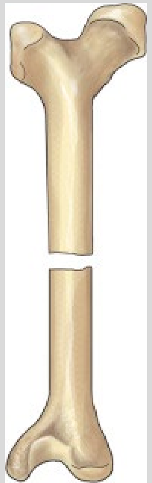
Fractures

Fracture Terminology:

- **Transverse** (A): fracture is perpendicular to the long axis of the bone
- **Oblique** (B): fracture down the long axis of the bone
- **Spiral** (C): a complex fracture line that encircles the shaft of the bone (twisting)
- **Comminuted** (D): more than 2 bone fragments
- **Avulsion** (E): fragment of bone torn away from the main mass of bone
- **Impacted** (F): bone fragments driven into each other, causing bones to wedge together

- **Nondisplaced**: fragments are aligned
- **Displaced** (G): fragments are separated
- **Compound**: displaced fracture that pierces the skin

A)



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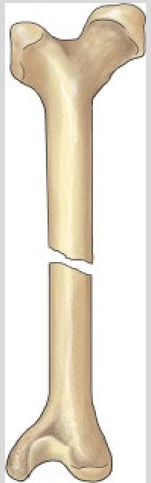

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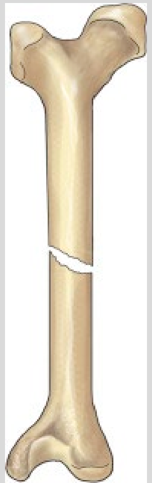

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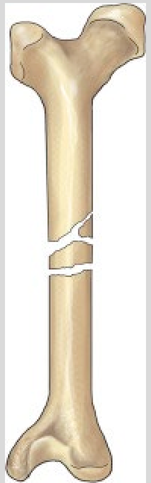

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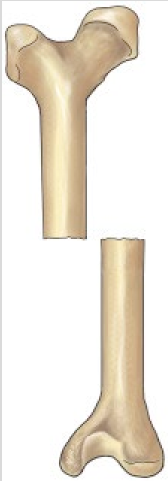

Sports Medicine - Real World Examples

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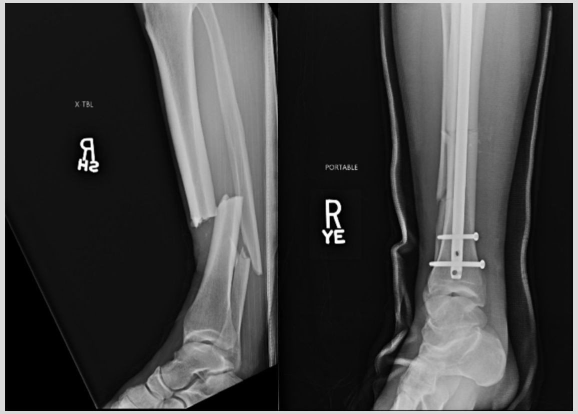
Chris Weidman, 202

https://twitter.com/chrisweidman



Screenshot from -
https://www.youtube.com/watch?v=mCPWm9Grh8Q&list=PLrdpldKEF236khwdFNRilNeOZded9gZLD&index=3

Alex Smith, 2018

https://www.espn.com/nfl/story/_/id/29112995/alex-smith-comeback-fight-save-qb-leg-life

Screenshots from -
https://www.youtube.com/watch?v=6bXO_AFowaY&list=PLrdpldKEF234DpxS6fOr9becIHtz7ibPt&index=85

Pediatric Fractures

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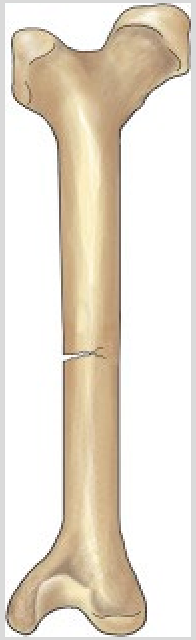
Fracture Terminology:

- **Greenstick** (A): fracture is complete except for a portion of the cortex on the **compression side** of the fracture
  - Incomplete fracture; usually due to a bending stress
  - Bone fails on **tension side**
- **Torus** (B): **buckling** of the cortex without breaking bone on the other side
  - Cortex buckles on compression and fractures on the **nontension side**

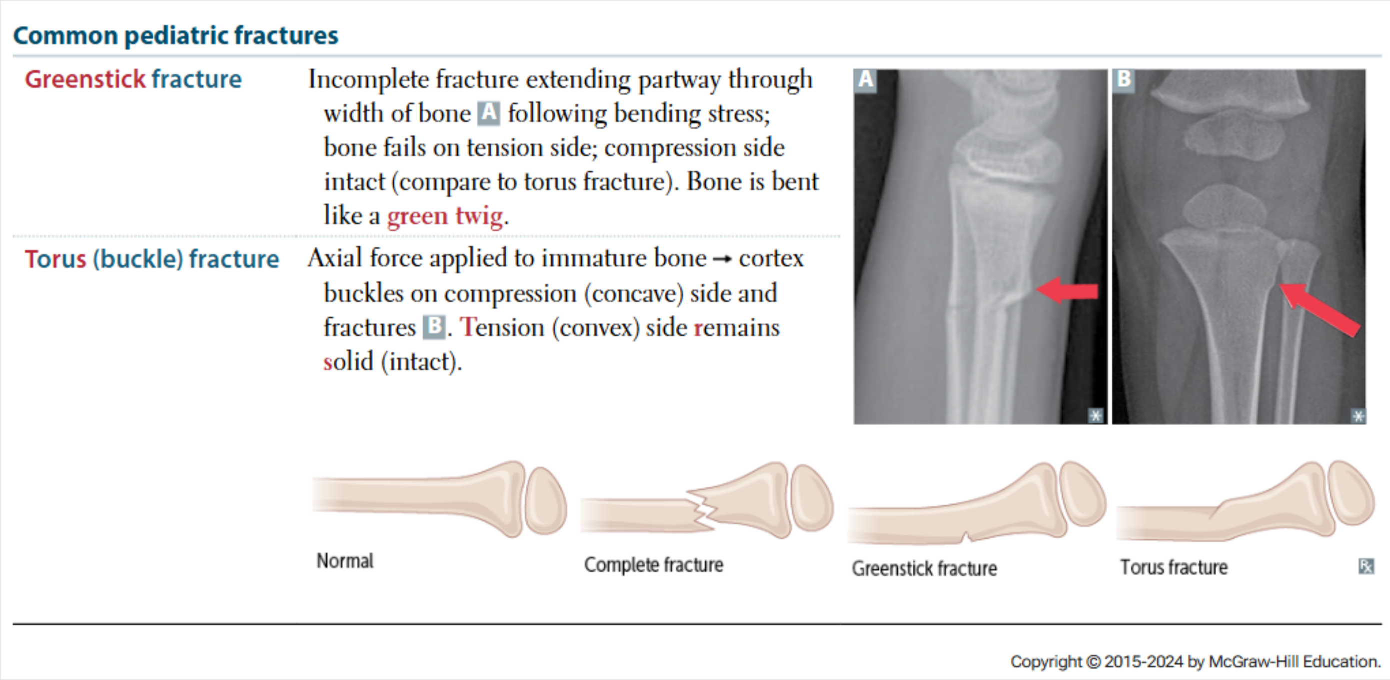
Greenstick



Essential Orthopaedics

Torus

Essential Orthopaedics



Common pediatric fractures

Greenstick fracture	Incomplete fracture extending partway through width of bone A following bending stress; bone fails on tension side; compression side intact (compare to torus fracture). Bone is bent like a green twig.
Torus (buckle) fracture	Axial force applied to immature bone → cortex buckles on compression (concave) side and fractures B. Tension (convex) side remains solid (intact).

Pediatric Fractures

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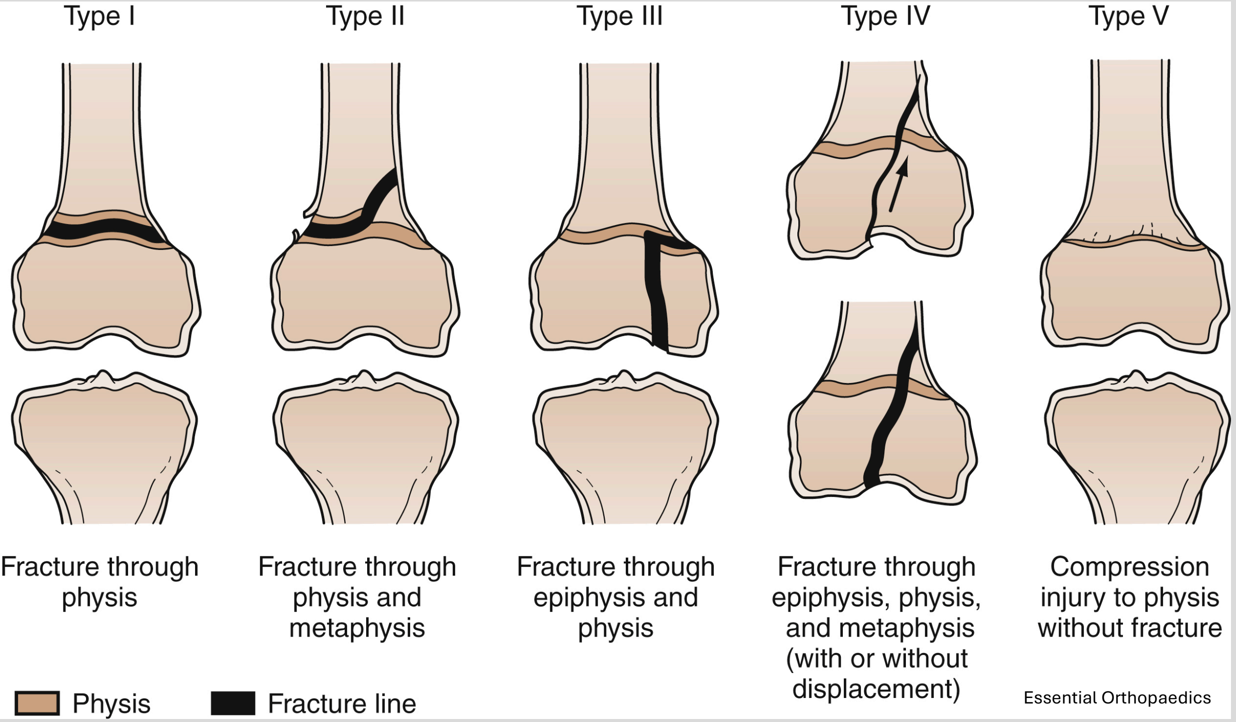
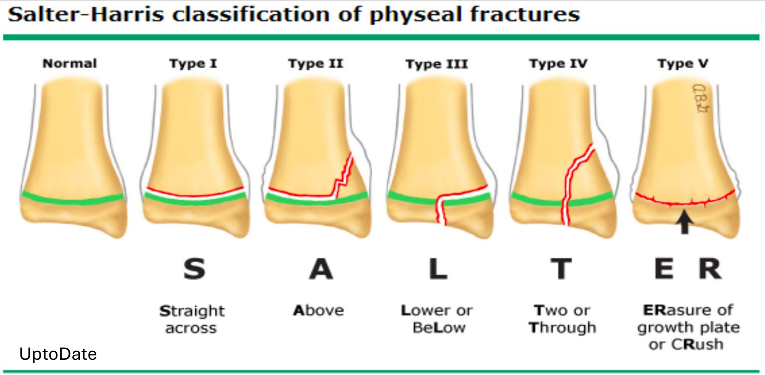
Salter-Harris Classification:

- A ranking/scheme used to classify physeal injuries





Type 1: fracture running through the growth plate itself

Type 2: fracture above the growth plate, through the metaphysis

Type 3: fracture lower or below the growth plate. Involves a horizontal fracture through the growth plate and vertical fracture of the epiphysis, creates a right angle

Type 4: fracture through the metaphysis, growth plate, and epiphysis vertically at an angle

- **Increased risk of growth arrest**

Type 5: crushing injury to the physis, no physeal fracture. No radiologic diagnosis can be made, only clinical

- **100% growth arrest rate (obliterated physeal plate)**

Compartment Syndrome

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- Skeletal muscles are contained within distinct osseofascial compartments throughout the body
- Compartment syndrome is due to increased pressure within these noncompliant compartments
    - **Acute Compartment Syndrome:** medical emergency, leading to permanent functional loss
        - Due to internal (fluid via hemorrhage/edema limiting space of neurovasculature) or external (compression or traction of limb changes compartment sizes)
    - Chronic Exertional Compartment Syndrome: usually exercise induced, will resolve w/ rest
- **Common Causes**:
    - Fractures (open and closed), most common being tibial diaphyseal fractures
    - Trauma/burns, constrictive bandages, animal bites/stings
- **Clinical Exam**: must take into account the entire picture
    - Pain: pain with passive movement **(sensitive sign)**
    - Pressure: increased tension of muscle compartments
    - Paresthesia: due to ischemia of the nerves
        - May be irreversible if compartment syndrome not treated
    - Paralysis: can be due to nerve/muscle injury OR inhibition due to pain
    - Pallor/Pulselessness: decreased capillary perfusion
- **Treatment**: fasciotomy

Bursitis

Bursa:
- A lubricated, fluid-filled sac adjacent or between soft tissues
- Purpose: reduce friction between structures, such as bone joints tendons and skin

Pathophysiology: bursae become irritated or inflamed, presenting acutely with pain and swelling
- **Chronic:** functional limitations due to joint contractures
- **Causes:** infection, chronic overuse, gout, rheumatoid arthritis

Presentation: pain, swelling, and tenderness near a joint
- **Acute**: pain and functional limitations over the course of a few days
- **Chronic:** long term symptoms that have been present for weeks to month
- May appear similar to tendinopathy:
  - In tendinopathy, pain is usually elicited with **movements** that place load or stretch on the tendon
- Common Areas:
  - Shoulder, patellar, olecranon, and Achilles

The Shoulder

Shoulder Anatomy

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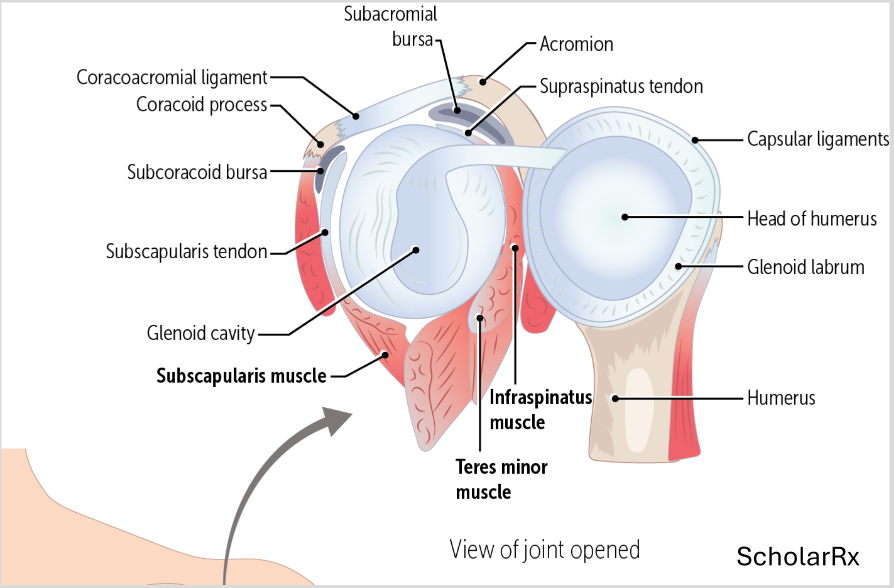
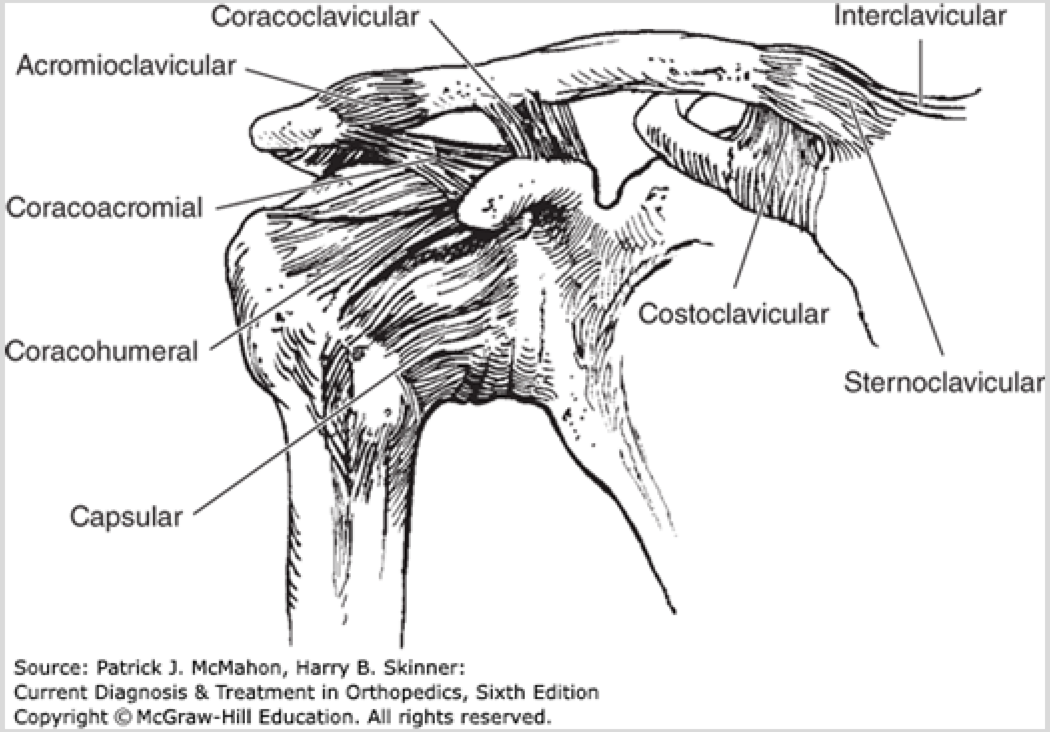
- Shoulder Girdle = Clavicle and scapula
- Sternoclavicular Joint (SC): only true joint connecting the shoulder girdle to the axial skeleton
    - Ball and socket joint type
    - Pro: increased mobility around the join
    - Con: increase instability/susceptibility to dislocations and sprains
- Acromioclavicular Joint (AC): only articulation between the clavicle and scapula
- Glenohumeral Joint: **greatest mobility of all joints**
    - Sacrifice stability for increased mobility
    - Held in place by the **labrum** (dense fibrous tissue)





Shoulder Anatomy

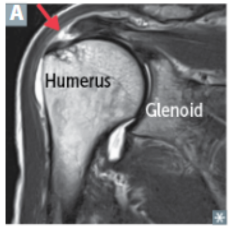
The Rotator Cuff:
- Primary active stabilizers of the GH joint, activation of these muscles compress the humeral head within the glenoid

- **Supraspinatus:** initiates shoulder abduction
- **Infraspinatus**: external rotator of the shoulder
- **Teres Minor**: external rotation and adduction
- **Subscapularis**: internal rotation

Shoulder Abduction:

Range of Motion	Muscle	Nerve
0°-15°	Supraspinatus	Suprascapular n
15°-90°	Deltoid	Axillary n
>90°	Trapezius	Accessory n (CN XI)
>90°	Serratus Anterior	Long Thoracic n

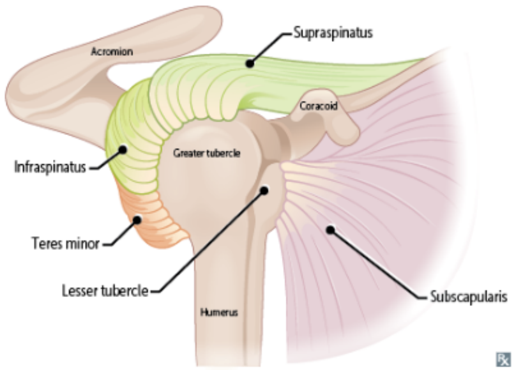
Rotator cuff muscles



Shoulder muscles that form the rotator cuff:
- Supraspinatus (suprascapular nerve)—abducts arm initially (before the action of the deltoid); most common rotator cuff injury (trauma or degeneration and impingement → tendinopathy or tear [arrow in A]), assessed by "empty/full can" test
- Infraspinatus (suprascapular nerve)—externally rotates arm; pitching injury
- teres minor (axillary nerve)—adducts and externally rotates arm
- Subscapularis (upper and lower subscapular nerves)—internally rotates and adducts arm

Innervated primarily by C5-C6.

SItS (small t is for teres minor).



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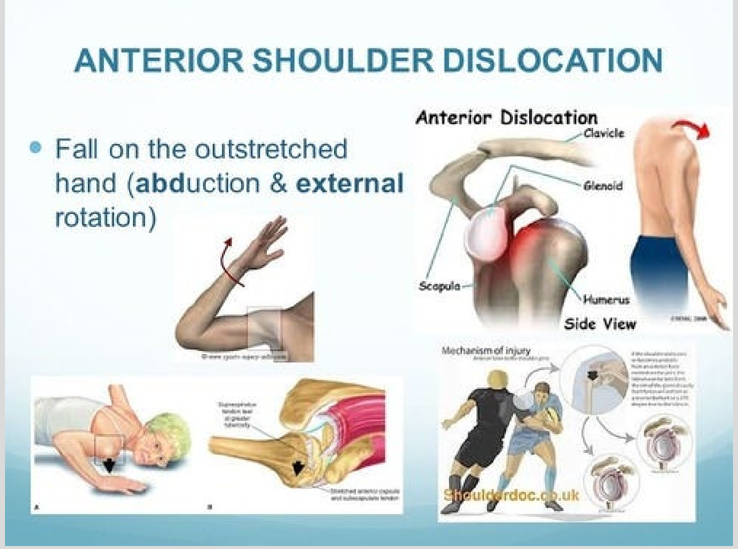
Shoulder Dislocations

- Shoulder Instability: excessive translation of the humerus over the glenoid surface
- Both static (bones & ligaments) and dynamic (muscles) stabilizers are needed to improve the stability of the GH joint
- Least stable shoulder positionn: **Abduction, External Rotation, Extension**

Anterior Dislocation: more common

- Common Mechanism: A blow to the medial aspect of the elbow while shoulder is abducted, extended, and externally rotated
- The head of the humerus will dislocate anteriorly from the GH joint
  - Mechanism: anterior capsule is stretched or torn
    - **Bankart Lesion**: anteroinferior capsular injury associated with a tear of the glenoid labrum off the anteroinferior glenoid limb
    - **Hill-Sachs Lesion**: compression fracture of the posterolateral humeral head
      - Occurs by the sharp edge of the anterior glenoid as the humeral head dislocates over it
    - **Both lesions increase the likelihood of recurrent dislocations**
- **Presentation:** held in slight abduction, and forearm is held by opposite hand, unable to be internally rotated. A dimple under the acromion could be seen
- **Treatment:** reduction of dislocated shoulder w/ immobilization for 2-6 weeks. Strength training, targeting rotator cuff stabilization, should be emphasized.
  - If these measures fail, surgical treatment can be considered



Shoulder Dislocations

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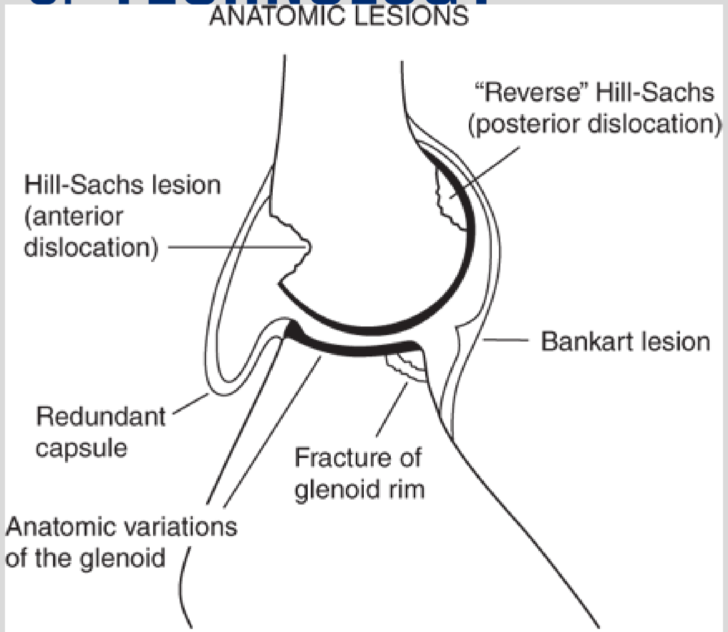
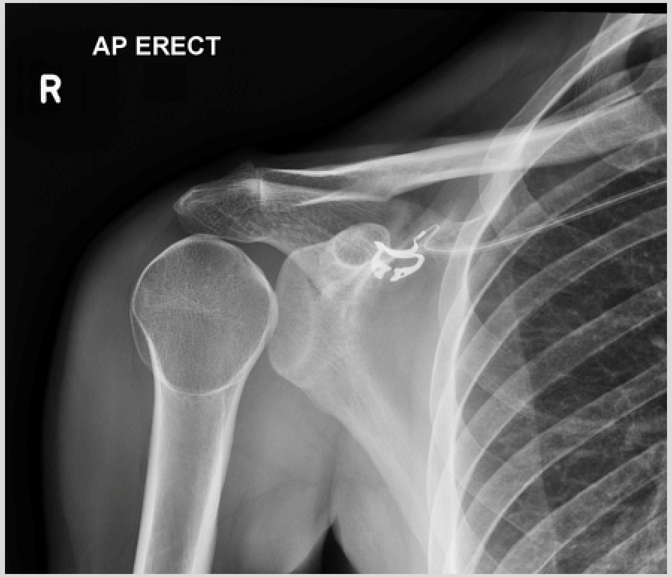


Source: Patrick J. McMahon, Harry B. Skinner: Current Diagnosis & Treatment in Orthopedics, Sixth Edition

Posterior:
- Rare, seen in long-term shoulder instability
- Commonly missed on imagining: light bulb sign
  - Humerus internally rotates with posterior dislocation, the GH joint widens
- Presentation: arm is typically held with the hand on the belly, cannot be externally rotated



https://radiopaedia.org/articles/posterior-shoulder-dislocation?lang=us

*PowerPoint edited from recorded version to add:
Posterior dislocation can be caused by forceful adduction with Internal Rotation, as seen in the classic injuries below::
1. Anterior to Posterior trauma (grabbing the dashboard in front of you in a car accident)
2. Seizure
3. Electrocution or being struck by lightning

Edited by MH

Labral Tears: Bankart Tear

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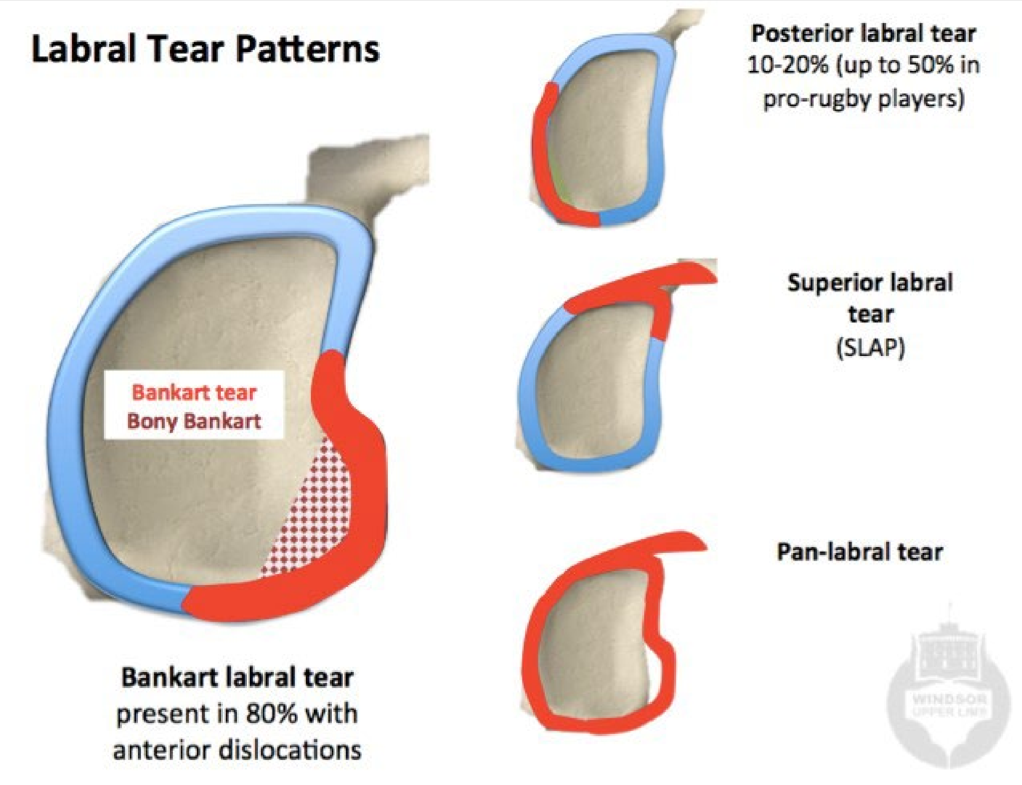
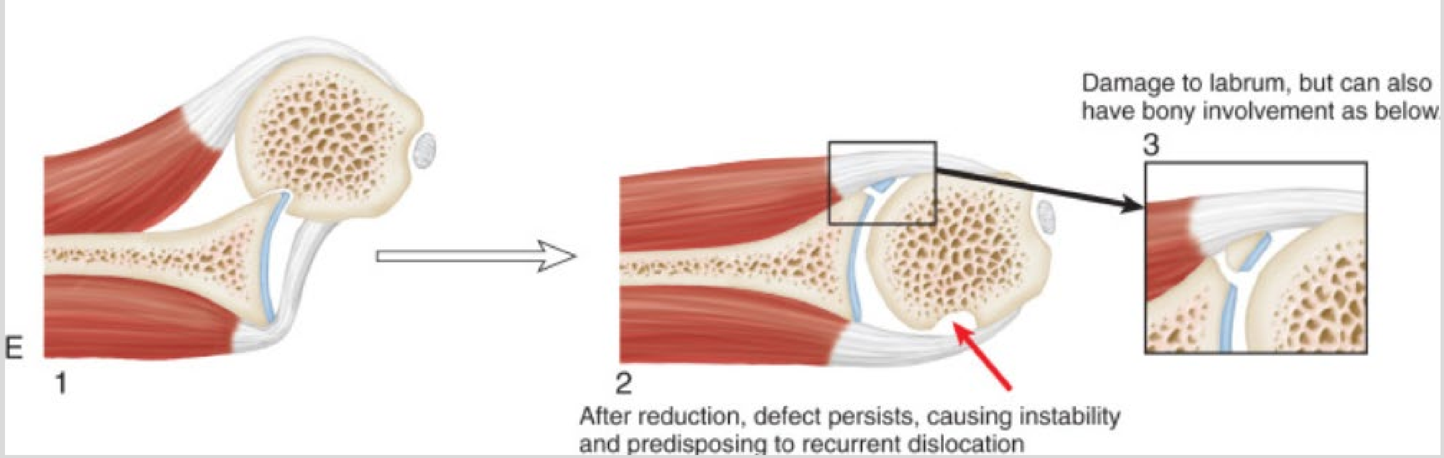
Bankart Tear

- **Tear to the inferior rim of the labrum**
- Mechanism: occurs with anterior dislocation
  - Avulsion of the anterior capsule-labral complex below the midline
  - Associated with Hill-Sachs lesion: injury to posterolateral aspect of humeral head
- Presentation:
  - Symptoms: popping, catching, grinding, and locking
- Associated with anterior shoulder dislocation and instability of the shoulder



https://www.windsorupperlimb.com/conditions/shoulder-conditions/shoulder-dislocation-instabilty-and-labral-tears



Labral Tears: SLAP Tear

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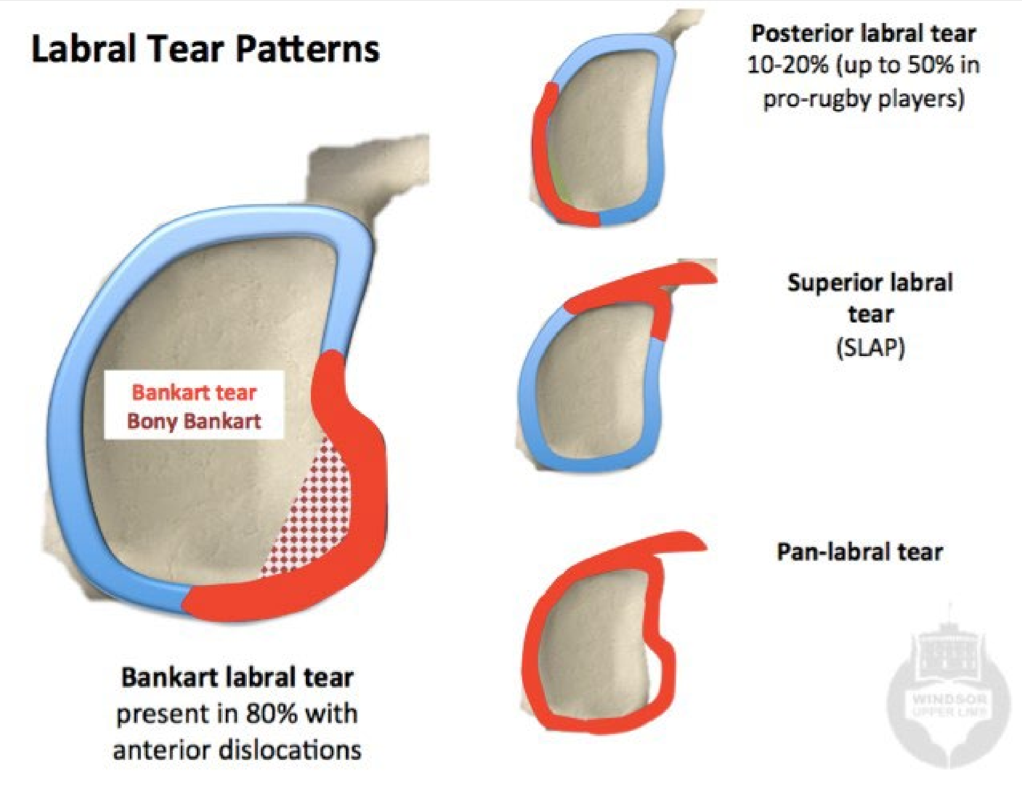
- Labrum is the cartilage lining the glenoid fossa, helping stabilize the shoulder
  - Increases surface area for humeral head contact and provides attachment for glenohumeral ligament and the long head of the bicep

Superior Labrum Anterior to Posterior (SLAP) Tear:

- Superior rim of the labrum
- Mechanism:
  - Acute Injury
  - Chronic Degradation: repetitive overhead motion (throwers, swimmers)
- **Presentation**: popping, catching, grinding and locking
  - Limited bicep contraction
    - Superior glenoid labrum acts as the attachment point for the long head of biceps
  - Special Tests: O'Brien Test, Yergason Test, Anterior Apprehension Test
- **Treatment:** ideal treatment is unclear
  - Typically, conservative treatments must fail prior to surgical intervention



https://www.windsorupperlimb.com/conditions/shoulder-conditions/shoulder-dislocation-instabilty-and-labral-tears

Acromioclavicular Dislocations

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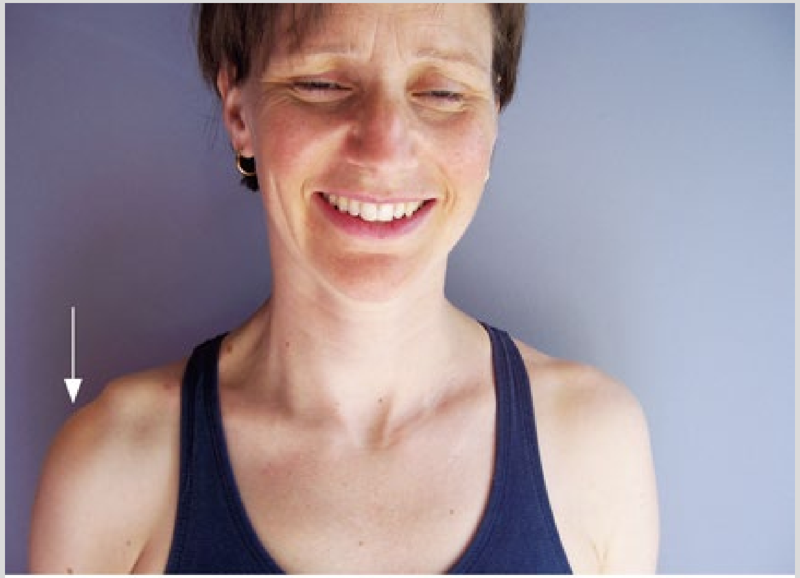
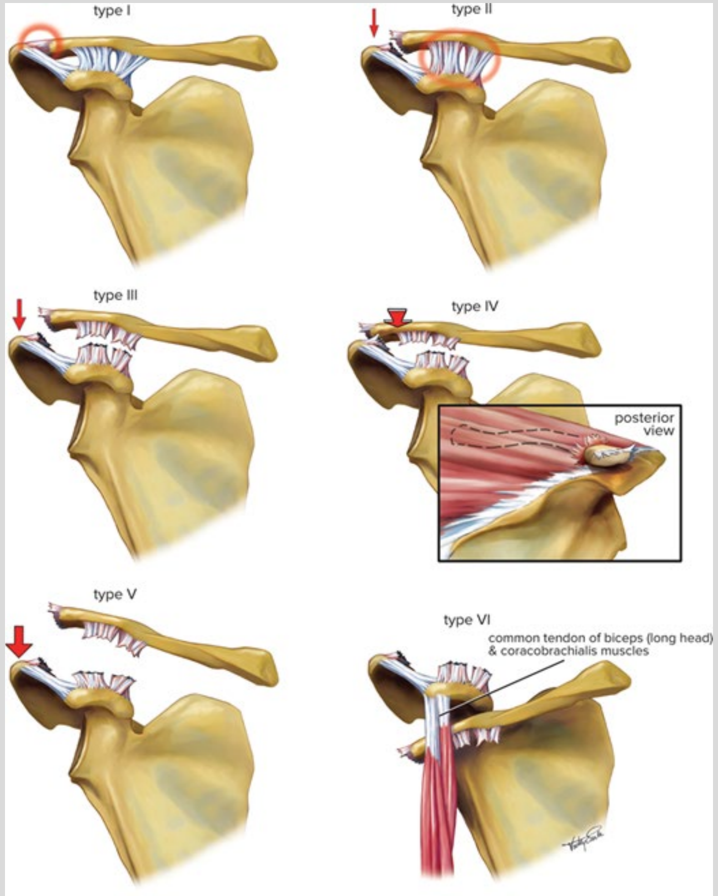
- Acromioclavicular (AC) joint is a common site of injury
  - Often due to trauma to shoulder area with an adducted arm
- In more severe dislocations, a marked step deformity will be present

Common Presentation:
- Pain
- Loss of function of the scapula
- Tenderness over the AC joint, with potential step off
- **Management:**
  - Ice and immobilization in a sling for pain relief
  - Duration of sling can vary from a few days to up to 6 weeks, depending on the severity of the dislocation



Source: Peter Brukner: *Brukner & Khan's Clinical Sports Medicine: Injuries, Volume 1*, 5e: www.csm.mhmedical.com

Essential Orthopaedics, 2e



Source: Peter Brukner: *Brukner & Khan's Clinical Sports Medicine: Injuries, Volume 1*, 5e: www.csm.mhmedical.com

Clavicle Fractures

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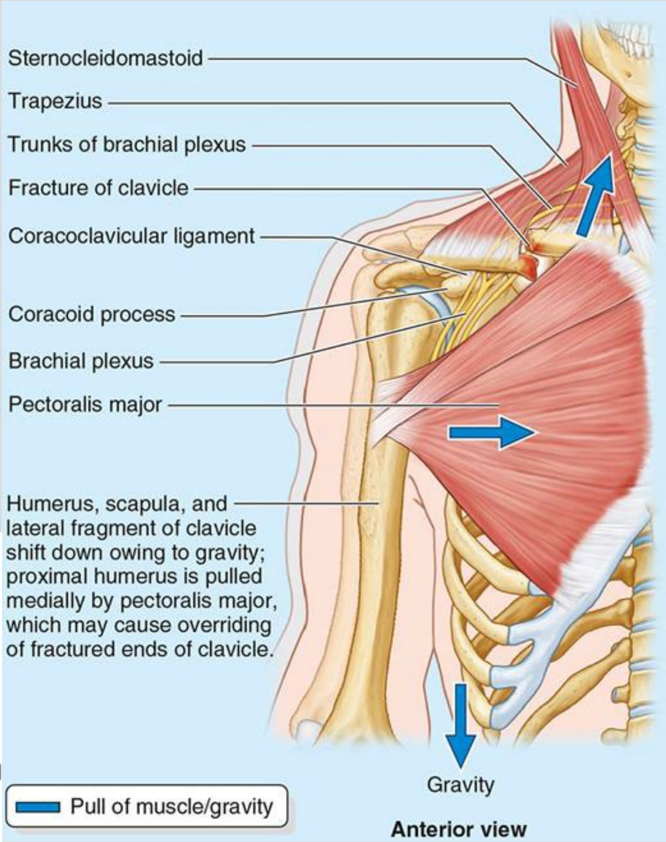
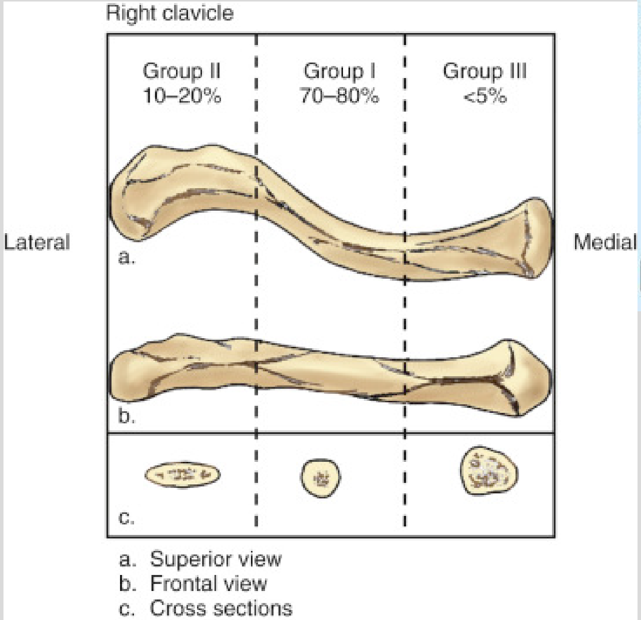
- Common in children and as birth trauma
- Clavicle fracture: location is broken into 3 sections (medial third, middle third (midshaft), and distal third
    - Commonly caused by **fall on outstretched hand or direct shoulder trauma**
    - Most common is **middle third (midshaft) fracture**
        - Presents with shoulder drop
            - Due to sternocleidomastoid muscle elevating the medial fragment of the clavicle
            - Weight of the shoulder pulls the lateral end of the clavicle down
        - Treatment: sling, most likely nonsurgical



Moore's Clinically Orientated Anatomy, 9e



Essential Orthopaedics, 2e

Rotator Cuff

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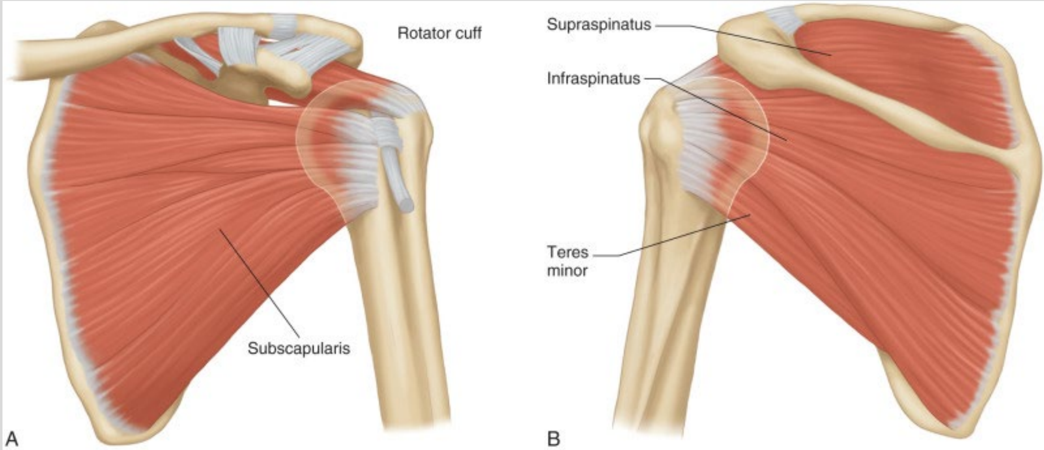
- **Supraspinatus** (suprascapular n): abducts the arm first 15°
  - ***Most common rotator cuff injury**
- **Infraspinatus** (suprascapular n): laterally rotates arm
- **Teres Minor** (axillary n): laterally rotates arm
- **Subscapularis** (upper & lower subscapular n): medially rotates arm

- Primarily innervated by C5-C6

- Attachments:
  - Supraspinatus, infraspinatus, teres minor: greater tubercle of the humerus
  - Subscapularis: lesser tubercle of the humerus

- Mechanism: result from acute trauma or due to chronic overuse and degeneration
  - More often seen in older population (>70 years old)
- Treatment:
  - Conservative management for partial tears (early surgery shows no benefit)



Anterior (A) and posterior (B) views of the shoulder.
Moore’s Clinically Oriented Anatomy, 9e

Rotator Cuff Tear

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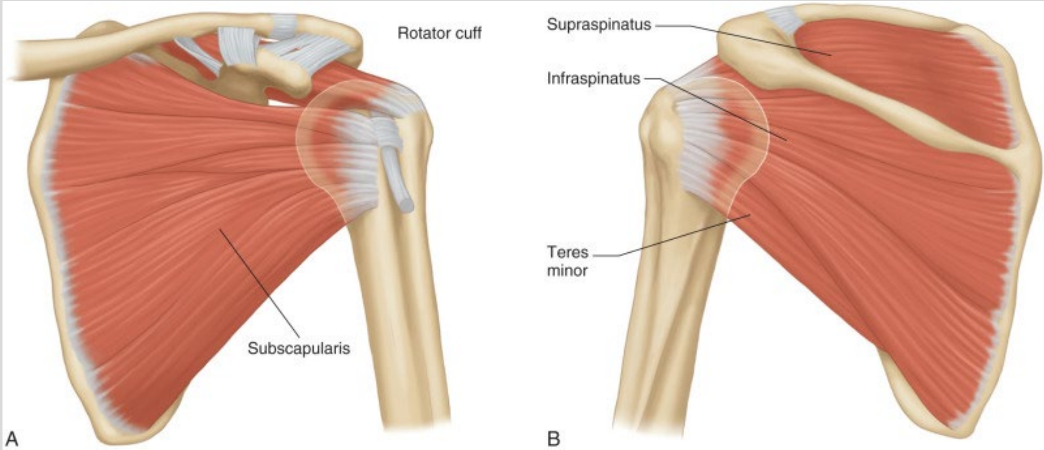
- **Mechanism**: result from acute trauma or due to chronic overuse and degeneration
  - More often seen in older population (>70 years old)
- **Presentation:**
  - Pain located over the anterior lateral shoulder
  - **Nighttime pain**
- **Physical Exam:**
  - Inspect for atrophy of supra/infraspinatus
  - Active and passive ROM testing
    - Can correlate to action of each muscle
  - Special tests: Empty can and Drop Arm test
- **Treatment**:
  - Conservative management for partial tears (early surgery shows no benefit)
  - Arthroscopic surgical intervention for full-thickness tears in younger patients
    - In elderly, conservative management is still effective



Anterior (A) and posterior (B) views of the shoulder.
Moore's Clinically Oriented Anatomy, 9e

Rotator Cuff Tear– Special Tests

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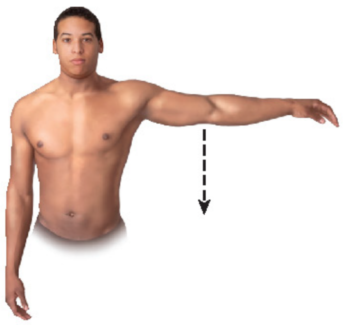
- **Empty Can:**
  - **Action:** Patient’s arm elevated to shoulder level in scapular plane and thumbs pointing down, examiner applies downward force and patient resists. Positive if weakness is present
  - **Tests**: supraspinatus muscle tear

- **Drop Arm Test:**
  - **Action:** Patient elevates arm fully and then slowly lowers arm. Positive if the arm suddenly drops or patient has severe pain
  - **Tests**: supraspinatus muscle tear

Bates' Pocket Guide to Physical Examination and History Taking, 9e

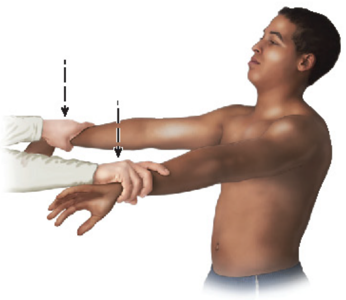
Strength Test

Drop-arm test. Ask the patient to fully abduct the arm to shoulder level, up to 90 degrees, and lower it slowly. Note that abduction above shoulder level, from 90 to 120 degrees, reflects action of the deltoid muscle.



Composite Test

Empty can test. Elevate the arms to 90 degrees and internally rotate the arms with the thumbs pointing down, as if emptying a can. Ask the patient to resist as you place downward pressure on the arms.



Impingement

- One of the most common causes of shoulder pain
- Includes pain caused by:
  - Rotator cuff tendinitis
  - Subacromial bursitis
  - Biceps tendinitis
- Mechanism: typically due to anatomic narrowing of the subacromial space
- Clinical Progression: edema, then fibrosis and tendonitis
  - History of repetitive overhead activity
  - Positive Neer and Hawkins sign
- Treatment
  - Mild disease treated conservatively (RICE)
  - Symptoms typically resolve
- Patients with chronic pain can benefit from cortisone injection around rotator cuff
  - Severe disease can require surgical repair
  - Removal of small portion of acromion or bursa can relieve pressure

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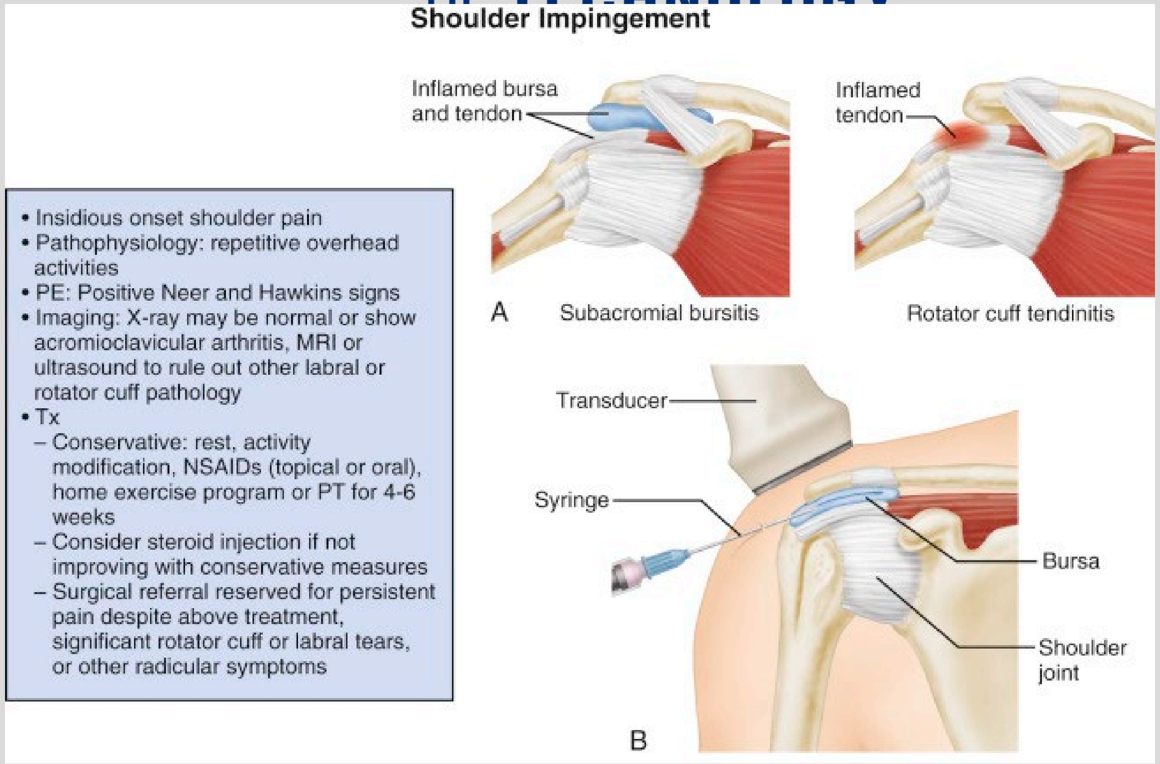


Essential Orthopaedics, 2e

Rotator Cuff Impingement - Special Tests

- **Hawkins' Test:**
  - **Action:** Examiner flexes the humerus and elbow to 90 degrees and then maximally internally rotates the shoulder and applies overpressure. Positive with reproduction of pain of the superior shoulder
  - Tests: subacromial impingement

- **Neer's Test:**
  - **Action:** Examiner stabilizes the scapula with a downward force while fully flexing the humerus overhead while applying overpressure. Positive with reproduction of pain of the superior shoulder
  - **Tests**: impingement

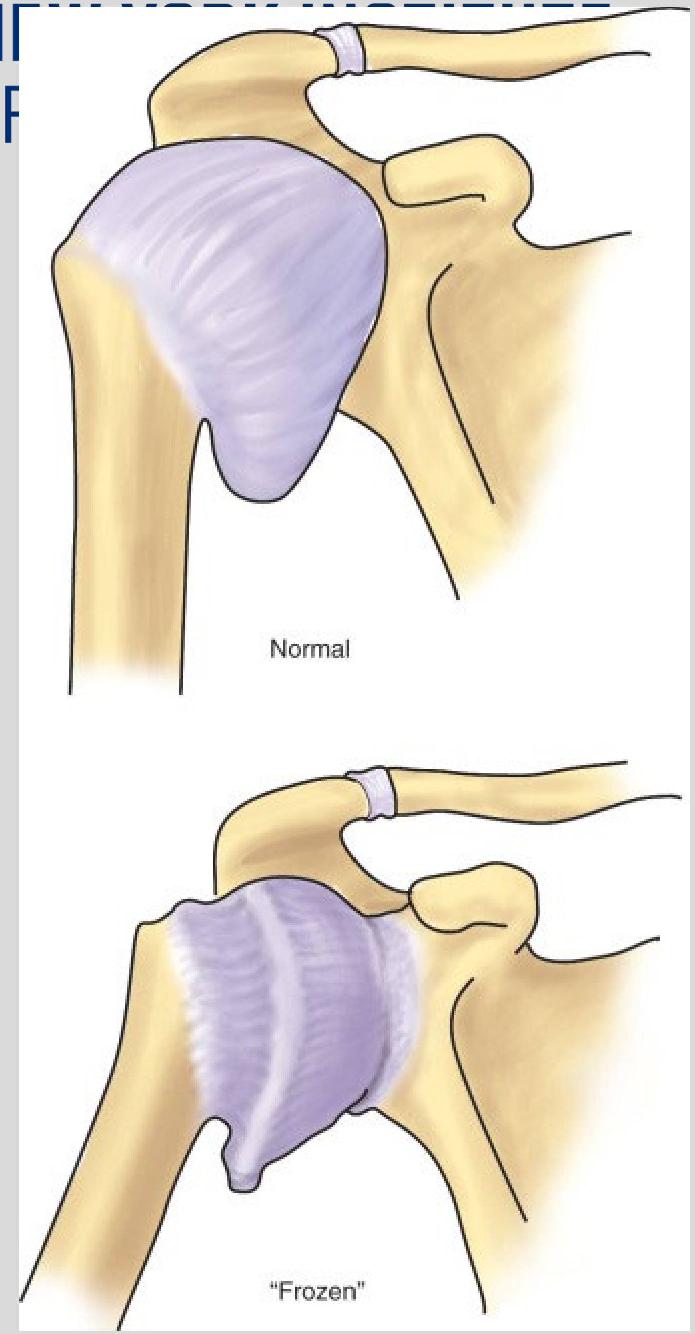
Bates' Pocket Guide to Physical Examination and History Taking, 9e

- *Hawkins impingement sign.* Flex the patient's shoulder and elbow to 90 degrees with the palm facing down. Then, with one hand on the forearm and one on the arm, rotate the arm internally. This compresses the greater tuberosity against the supraspinatus tendon and coracoacromial ligament.

- *Neer impingement sign.* Press on the scapula to prevent scapular motion with one hand and raise the patient's arm with the other. This compresses the greater tuberosity of the humerus against the acromion.

Adhesive Capsulitis

- Pathophysiology: poorly understood
  - Fibrosis, inflammation, and capsular contracture of the shoulder capsule
  - Primary is typically idiopathic, secondary is often due to a variety of underlying conditions and risk factors
  - **Responsible for pathologic loss of motion**
- Tends to affect middle aged patients in their non-dominant arm
- Commonly associated with diabetes mellitus
- **Presentation:**
  - Pain on palpation of anterior and posterior capsule
  - Active and passive loss of ROM
- Treatment:
  - Physical Therapy
  - Oral medications



Essential Orthopaedics, 2e

The Humerus

Fractures of the Humerus

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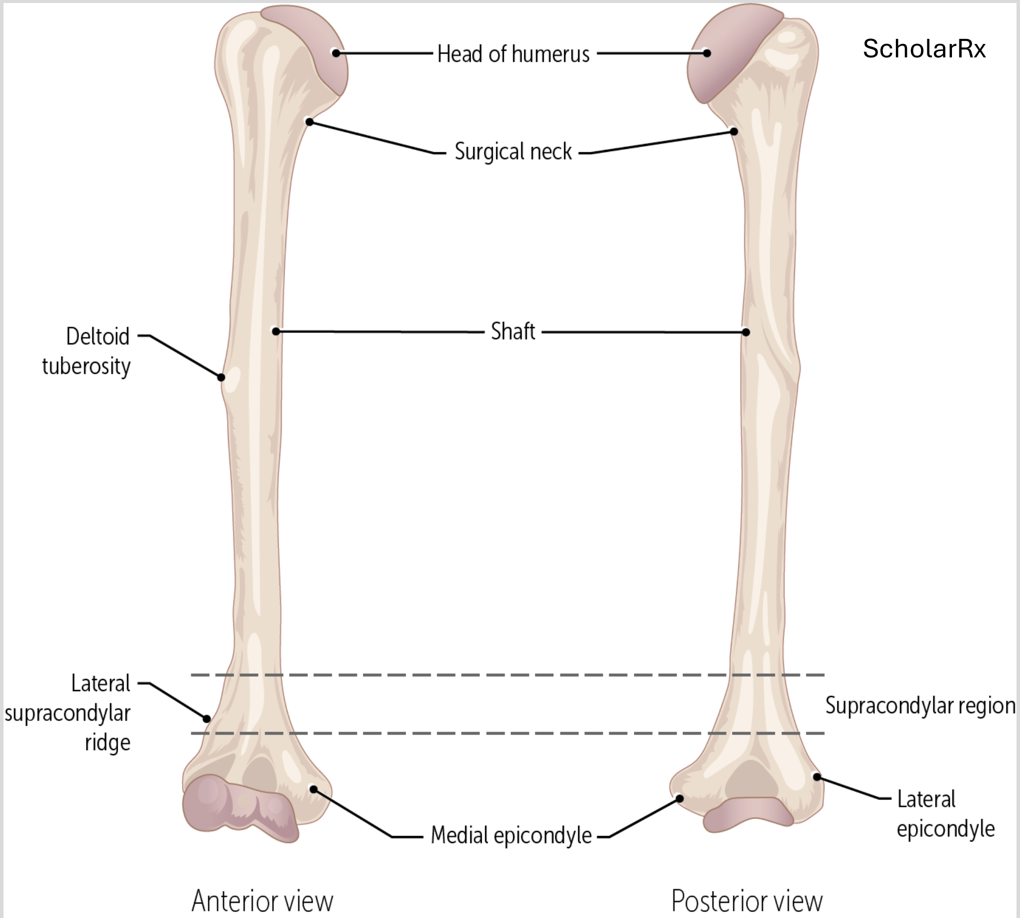
- Mechanism:
  - Low energy fall in osteoporotic/elderly individuals
  - High Energy trauma in young individuals

- Treatment: surgical or non-surgical

Keep in Mind

- **Which nerve is affected by this fracture**
- **What vascular structures are compromised by the fracture**
- **What is the clinical picture associated with each fracture type**



Surgical Neck Fracture of Humerus

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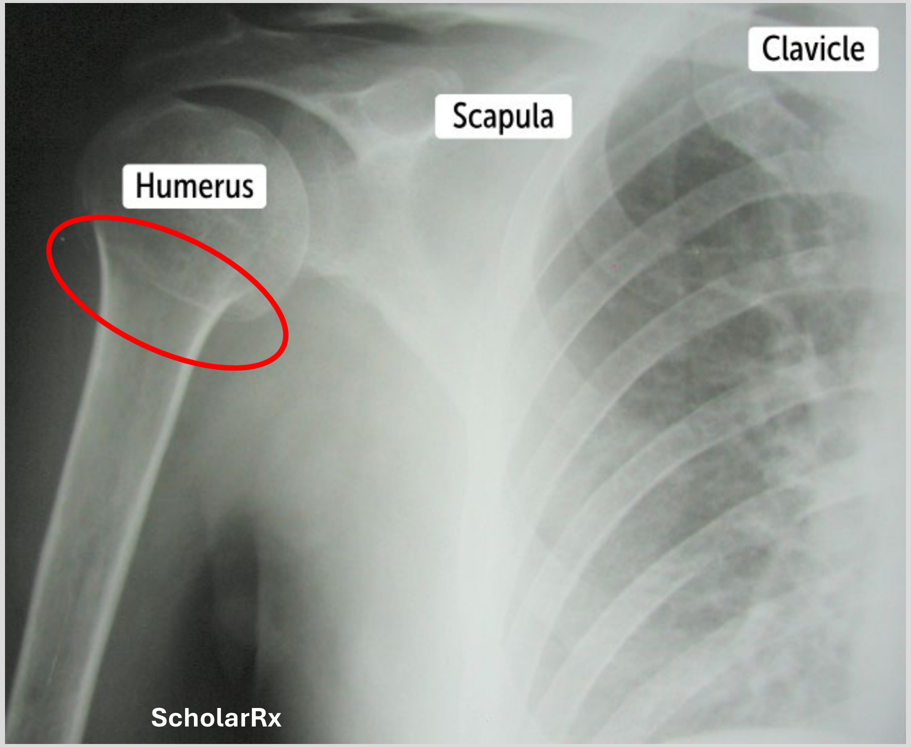
- Mechanism:
  - Most common: elderly due to mechanical fall
  - Direct blow or violent muscle contraction
  - Anterior dislocation of the humeral head

- Damage:
  - **Nerve**: axillary nerve
  - **Vasculature**: posterior circumflex humeral artery



Presentation:
- Flattened deltoid (axillary n)
- Loss of arm abduction at shoulder (>15° of abduction); (axillary n)
- Loss of sensation over deltoid and lateral arm (axillary n)

Midshaft Humeral Fracture

- Mechanism: direct blow, fall, collision
- Damage:
  - **Nerve**: radial nerve
  - **Vasculature**: deep brachial artery

Presentation:

- Loss of sensation over posterior arm/forearm and dorsal hand (radial n)
- Wrist drop, no extension of elbow, wrist, and finger (radial n)
  - Decreased grip strength

- This presentation is similar to compression of the radial nerve as it exits the brachial plexus (Saturday night palsy)

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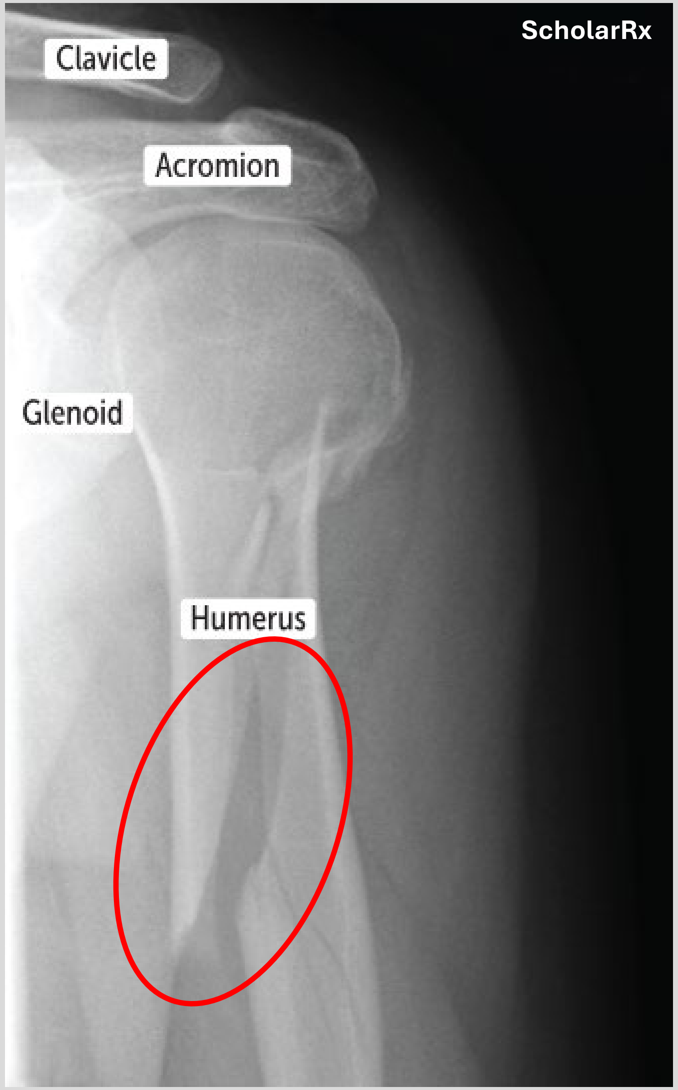

Distal Humerus (Supracondylar) Fracture

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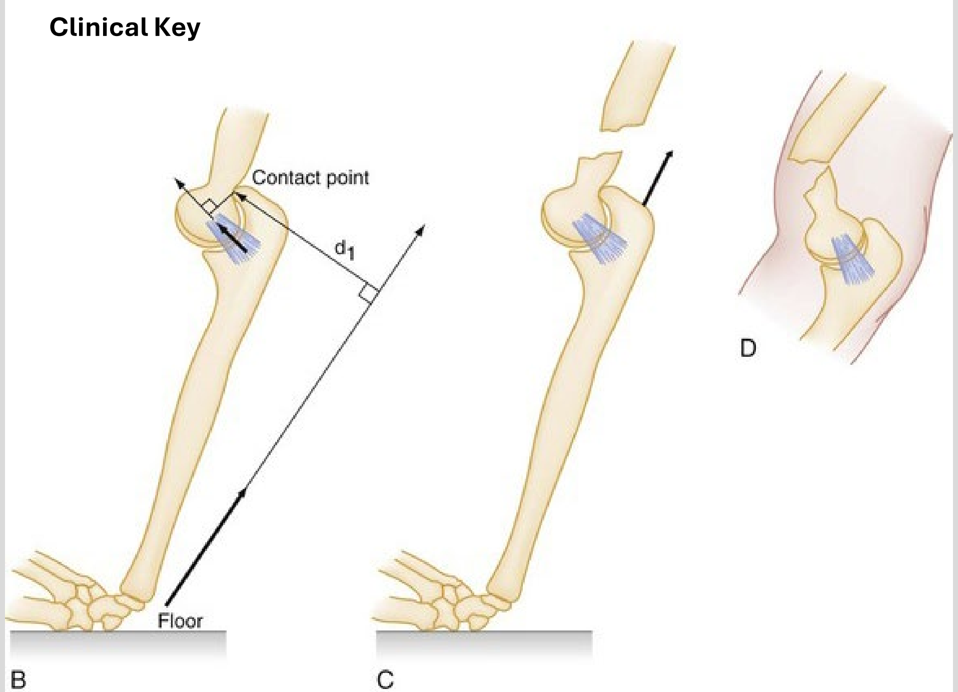
- Mechanism: most common in pediatric patients
  - FOOSH (Fall on Outstretched Hand) with extension in arm

- Damage:
  - **Nerve**: median nerve
  - **Vasculature**: brachial artery

Presentation:

- Loss of wrist flexion and functions of the lateral two lumbricals, Opponens pollicis, abductor pollicis brevis
  - Median n: flexor compartment of wrist
  - Similar to a carpal tunnel presentation



Orthobullets

Complications: Volkmann's Contracture

- Pathophysiology: due to the irreversible ischemia and compartment syndrome
  - Can result after a supracondylar humeral fracture
  - The swelling caused by the fracture leads to compartment syndrome, suppressing the brachial artery, leading to ischemia of the forearm flexor muscles
  - Additionally, median nerve damage further exacerbates damage and contraction to the flexor compartment
  - Overtime, the muscles shorten and scar, contributing to the contracture

- **Clinical Presentation**: fingers and wrist flexed, with passive extension nearly impossible

Medial Epicondyle Fracture

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- Mechanism: most common in pediatric patients
  - Usually a direct trauma

- Damage:
  - **Nerve**: ulnar nerve

Presentation:

- Loss of the ulnar nerve function:
  - Paresthesia in the 5th digit and medial half of 4th digit
  - Decreased grip strength (loss of flexion in digits 4-5)
  - Similar to **cubital tunnel syndrome**

https://www.drahmadsportsmedicine.com/expertise/elbow/medial-epicondyle-avulsion-fractures/

https://ufhealth.org/conditions-and-treatments/cubital-tunnel-syndrome

Humeral Fracture: Recap

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- **Surgical Neck**: Nerve: Axillary n (C5-C6) Artery: Posterior Circumflex Humeral a
  - Mechanism: anterior dislocation of shoulder/fall
  - **Presentation**:
    - Muscle: loss of arm abduction >15°
    - Sensory: loss of sensation over deltoid and lateral arm
- **Midshaft: Nerve**: Radial n (C5-T1) Artery: Deep Brachial
  - Mechanism: nonspecific
  - **Presentation**:
    - Muscle: loss of elbow, wrist, finger **extension**. Decreased grip strength
    - Sensory: loss over posterior arm-forearm and dorsal hand
- **Supracondylar**: Nerve: Median (C5-T1) Artery : Brachial
  - Mechanism: FOOSH
  - Presentation:
    - Muscle: loss of wrist **flexion**, thumb opposition
    - Sensation: loss over thenar eminence, dorsal and palmar aspects of lateral 3 ½ fingers
- **Medial Epicondyle**: Nerve: Ulnar (C8-T1)
  - Mechanism: direct trauma
  - Presentation:
    - Muscle: radial deviation of wrist w/ flexion, decreased flexion of 4^{th} and 5^{th} digits
    - Sensation: loss over 5^{th} and medial 4^{th} digit

Humeral Fracture: Recap

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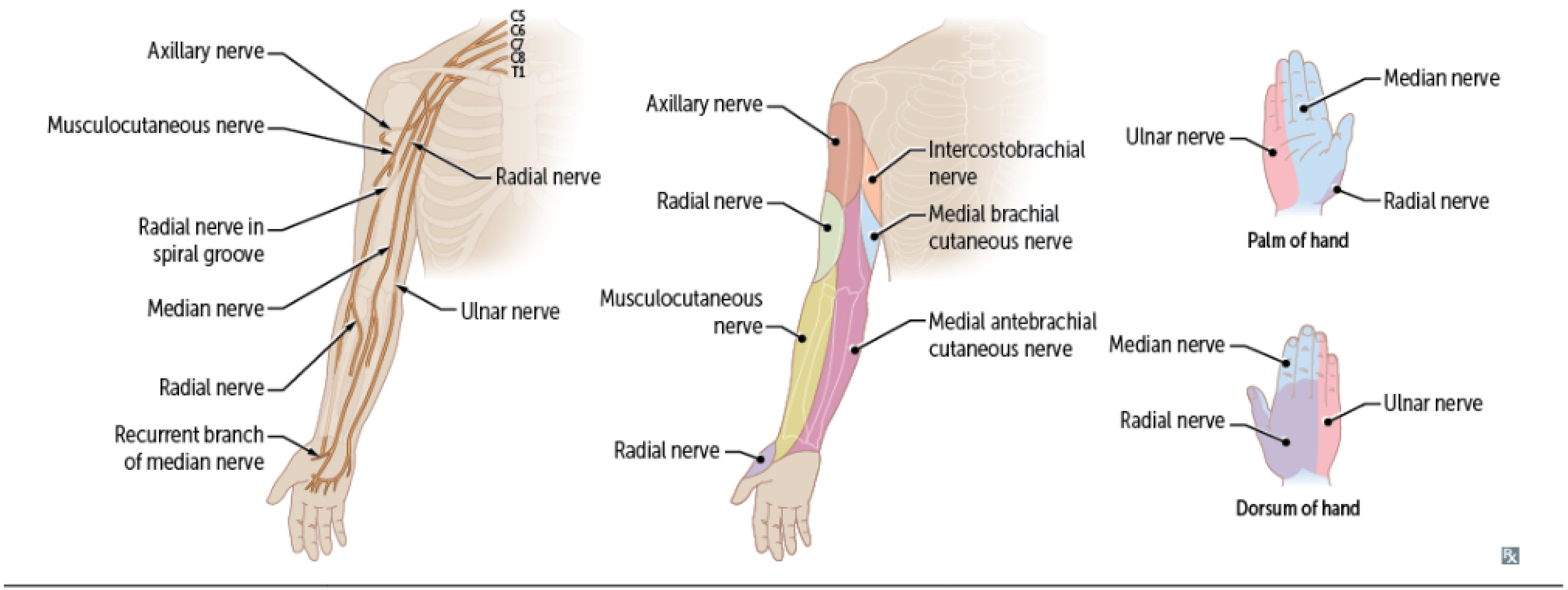
[partially hidden] LOGY



LOCATION	NERVE	ARTERY
Axilla/lateral thorax	Long thoracic	Lateral thoracic
Surgical neck of humerus	Axillary	Posterior circumflex
Midshaft of humerus	Radial	Deep brachial
Distal humerus/cubital fossa	Median	Brachial

The Elbow

Elbow Fractures:

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- Identifying elbow fractures is usually done with the help of the posterior fat pad

Posterior Fat Pad:

- Normally pressed in the olecranon fossa by the triceps tendon, making it invisible on lateral radiograph of the elbow
- When there is a fracture involving the elbow (distal humerus or radial head), inflammation and effusion develops
  - Increased fluids force the fat pad out of its normal physiologic resting place
- This movement creates a visible "posterior fat pad sign"
- The sign is often the **only visible** marker of a fracture, particularly in the pediatric population

http://radiologyinthai.blogspot.com/2011/04/fat-pad-sign.html

Elbow Dislocation:

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- **Mechanism:** trauma
    - Anterior Dislocation: typically an anterior to posterior force
    - Posterior: FOOSH
- Two Types:
    - Simple (no underlying fractures) and complex (associated fractures)
- Can have associated ligament tears, fractures, and ulnar nerve injury

Moore’s Clinically Oriented Anatomy, 9e

https://litfl.com/elbow-dislocation/

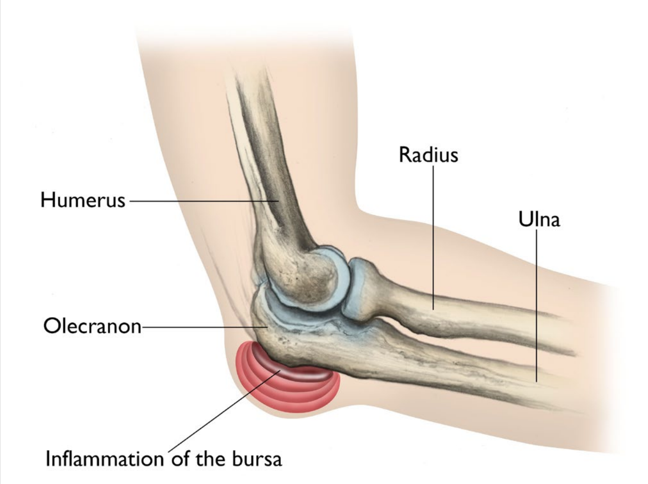
Dr. M. Heller, 2021

Olecranon Bursitis:

- Presentation can look similar to elbow dislocation, however it is important to remember the mechanism

- Mechanism: direct trauma or repeated excessive pressure/friction
  - Common in students, athletes, mechanics, plumbers
- Treatment: RICE and avoidance of aggravating activity
  - Drainage of the bursa is common
  - More aggressive treatments such as corticosteroid injections
  - If conservative measures fail, surgical removal of the bursa
- If there are concerns for septic bursitis, cultures from the drainage will be performed

Moore's Clinically Oriented Anatomy, 9e



Orthoinfo

Radial Head Subluxation

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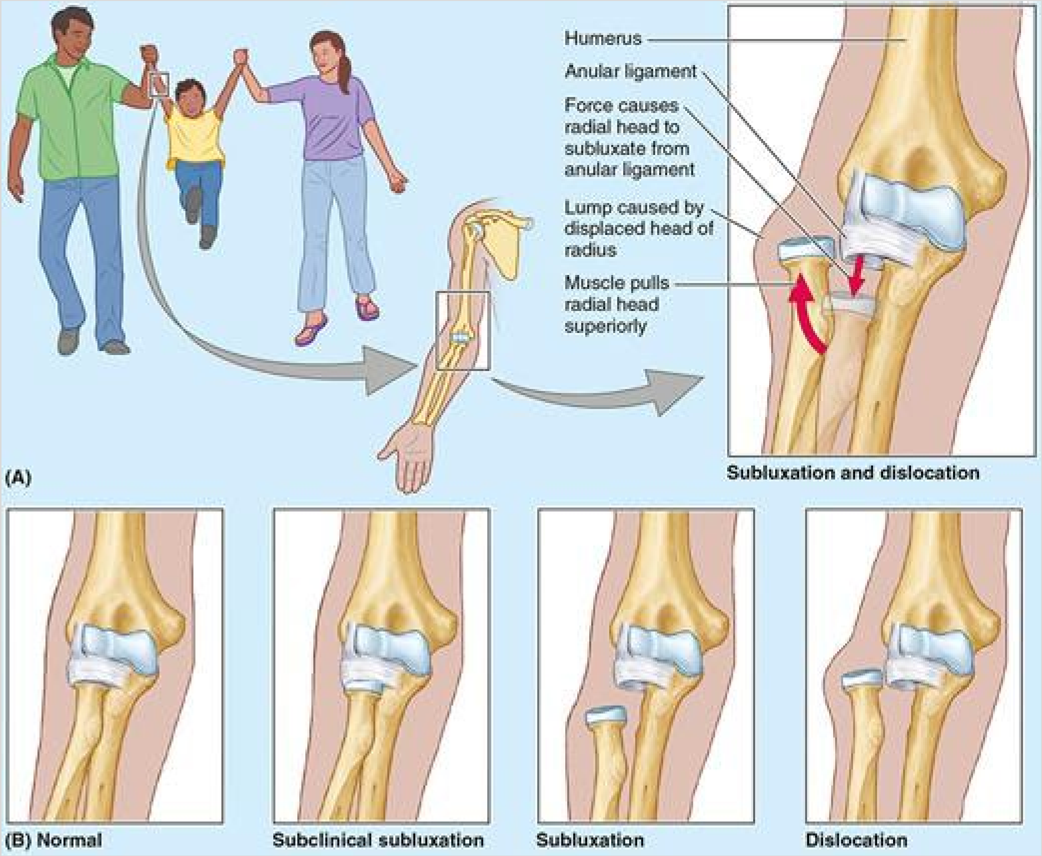
- Also known as nursemaid's elbow

- Mechanism: pulling a child by the hand or wrist
    - Force causes the radial head to displace inferiorly from the annular ligament
    - The **annular ligament** is immature, and will slip over the head of the radius
- Presentation: child will hold their hand in **flexion, pronation, and adduction**
- Treatment: radial head reduction
    - Performed with marked supination and flexion, audible pop will be heard
    - Immediate improvement in ROM will be seen once radial head is reduced



Moore's Clinically Oriented Anatomy, 9e

UCL Injury

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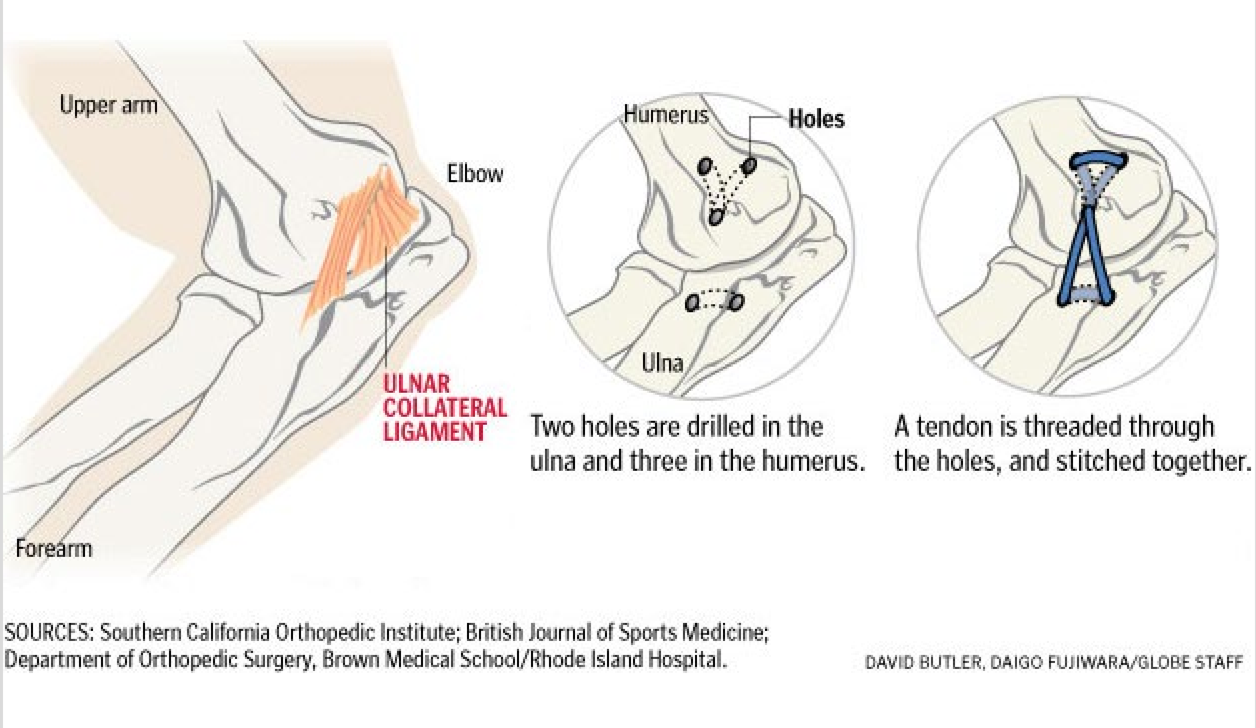
- Ulnar Collateral Ligament: attaches the humerus to the ulna
  - Three bundles: anterior, posterior, and oblique
- Mechanism: typically a valgus force at 50-90° of flexion puts the most stress on the UCL
  - This force is often seen at the acceleration phase of throwing
  - Repeated valgus stress on the medial elbow can break down the ligament
- Treatment: Tommy John Surgery

ESPN

https://caringmedical.com/prolotherapy-news/tommy-john-surgery-research-reveals-surgery-realities-success/



The Forearm

Epicondylitis

Tennis on the Outside, Golf on the Inside
MEGA: Medial Epicondylitis (Golfer's Elbow)
LATE: Lateral Epicondylitis (Tennis Elbow)

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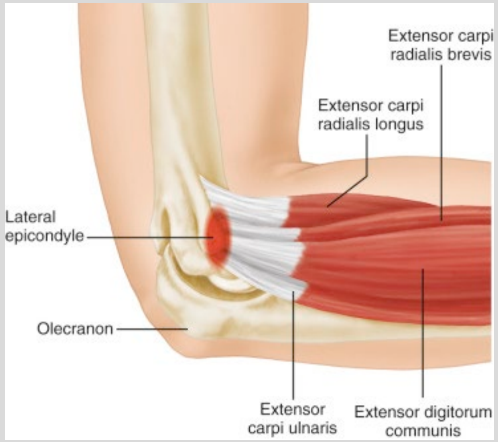
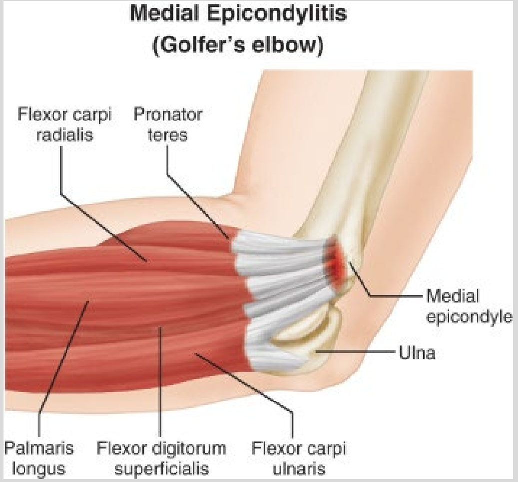
Lateral Epicondylitis: Tennis Elbow; Extension
- **Mechanism:** associated with chronic overuse of finger and forearm extension
  - Causes tendinosis of the common extensor tendon, at the lateral epicondyle
- Muscle group is innervated by the **Radial n**
- Physical Exam: palpate lateral epicondyle while passive pronation of the arm and wrist flexion OR resist active wrist extension
  - Positive test: pain near **lateral epicondyle**



Essential Orthopaedics, 2e

Medial Epicondylitis: Golfer's Elbow; Flexion & Pronation
- **Mechanism:** associated with chronic overuse of wrist flexors and forearm pronators
  - Causes tendinosis of the common flexor tendon at the medial epicondyle
- Muscle is innervated by the **Median n**
- Physical Exam: palpate medial epicondyle with passive supination of arm, extension of the wrist OR resist active wrist flexion
  - Positive test: pain near **medial epicondyle**



Distal Radius Fracture

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- Most common injury in orthopedics
- Mechanism: FOOSH
  - Higher risk in elderly (osteoporosis)

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Cubital Tunnel Syndrome

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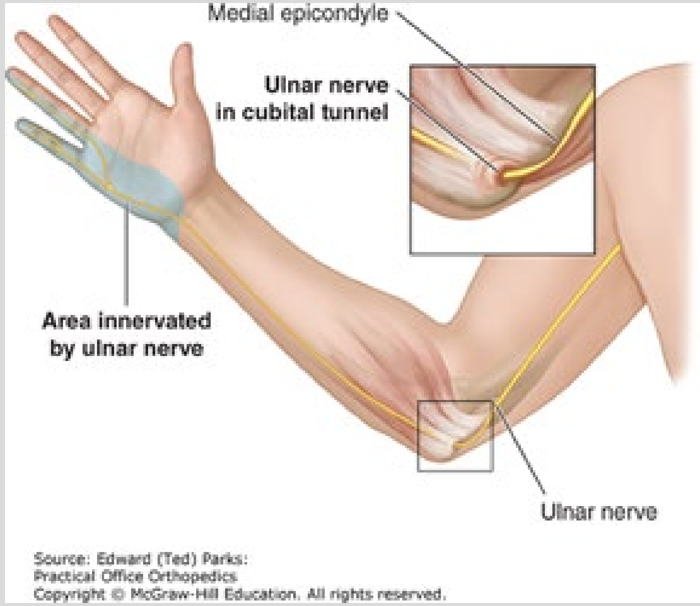
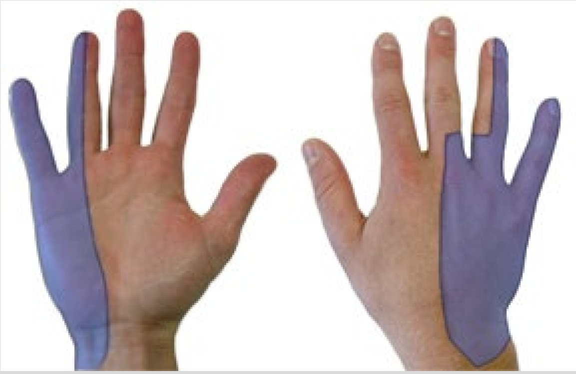
- **Mechanism**: ulnar nerve is entrapped at the elbow within the cubital tunnel
  - Cubital tunnel is a 2-3-inch-long tunnel near the medial epicondyle
  - The ulnar nerve is compressed by rigid connective tissue
- **Physical Exam:**
  - Tapping over the cubital tunnel will create electrical shooting sensation into the ulnar distribution of the hand (digits 4-5)
  - Paresthesia of digits 4-5
  - **Weakness of intrinsic hand muscles** and **inability** to **flex digits** 4-5 due to a weak flexor digitorum profundus





https://www.orthobullets.com/hand/6021/cubital-tunnel-syndrome?hideLeftMenu=true

The Wrist

Scaphoid Fracture

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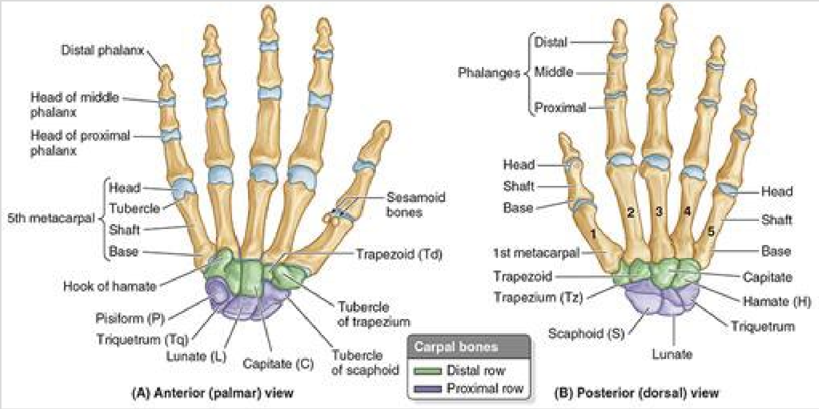
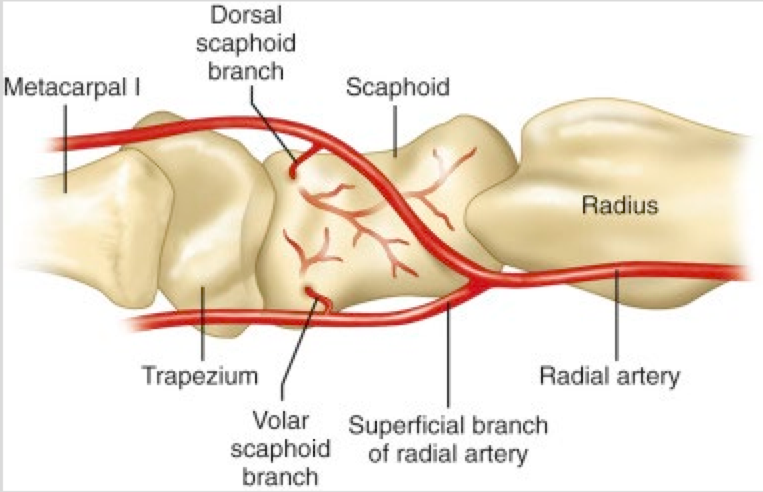
- 8 carpal bones (Proximal to distal, lateral to medial then medial to lateral)
  - **S**o **L**ong **T**oo **P**inky, **H**ere **C**omes **T**he **T**humb

- The **Scaphoid** is the most commonly fractured carpal bone
  - Mechanism: **fall on an outstretched hand**
  - Palpable in the anatomic snuffbox
  - With fracture, there is risk of avascular necrosis and nonunion
    - Scaphoid is supplied with a retrograde blood supply from a branch of the radial artery
    - A more **proximal injury increases the risk of necrosis**
  - Physical Exam: painful and limited ROM of the wrist, with **pain on palpation of the anatomic snuffbox**
  - Treatment:
    - If no risk of avascular necrosis (more distal fracture), casting in a spica cast for up to 20 weeks
    - If increased risk of necrosis or accelerated return to play timeline: surgery

  - Clinical Pearl: X-Ray has a lower sensitivity for identifying scaphoid fractures, a CT or MRI may be necessary to identify





Essential Orthopaedics, 2e

Scaphoid Fracture

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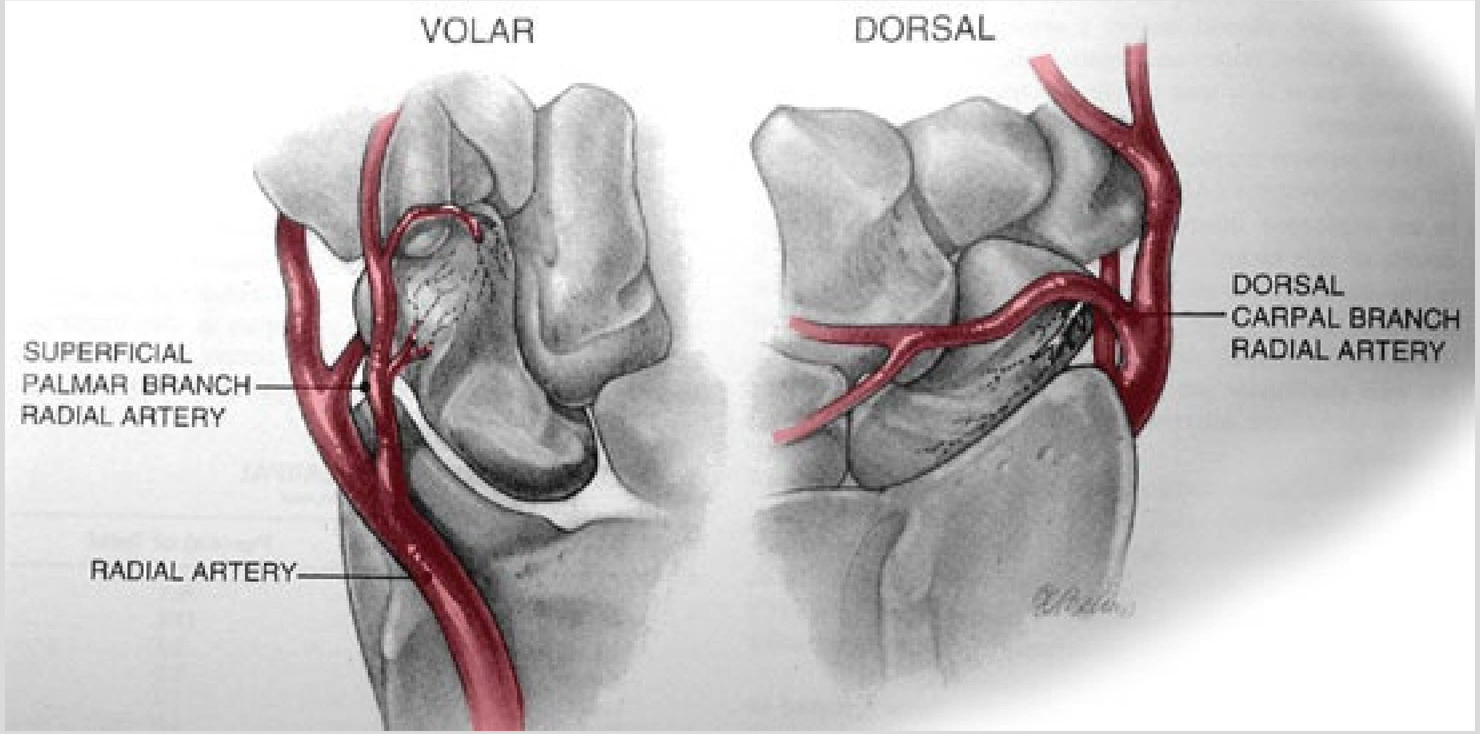
- Often missed on initial X-ray! (Repeat imaging commonly required)



https://www.cureus.com/articles/37247-conservatively-managed-displaced-scaphoid-fracture-in-a-young-female-collegiate-basketball-player

https://www.orthobullets.com/hand/6034/scaphoid-fracture

Metacarpal Neck Fracture

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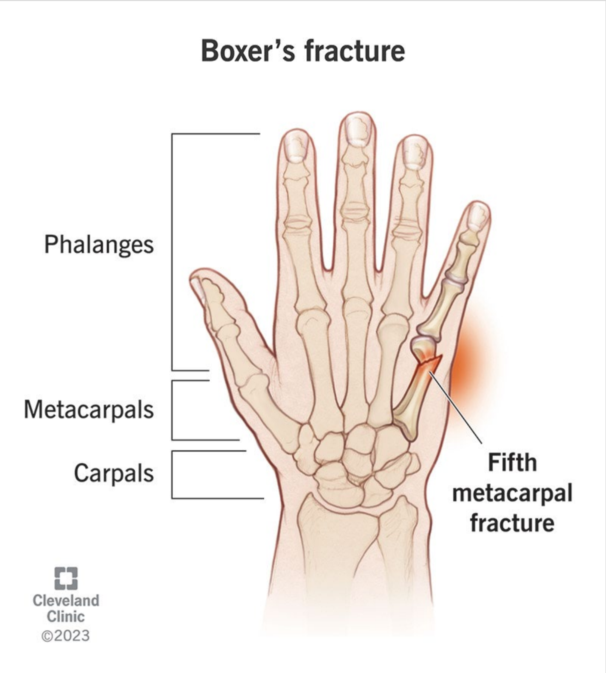
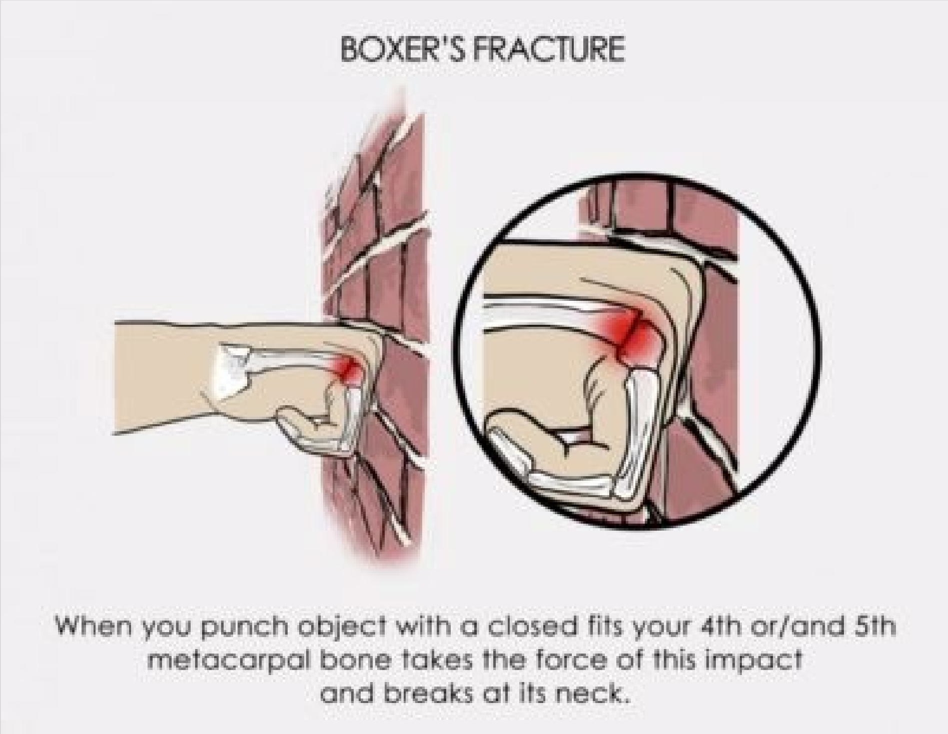
- Most common hand injury

- Mechanism: direct blow to hand (boxers' fracture)
- Physical Exam: deformity at metacarpal base, potential shortening of the fracture metacarpal
    - Typically no motor deficits





https://www.sportsmd.com/sports-injuries/wrist-hand-injuries/boxers-fracture/

Fight Bite

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- Typically seen in a clenched fist injury, where a struck tooth ruptures the metacarpophalangeal joint (MCP), invading it with mouth organisms
  - These cuts can be missed without thorough physical exam
- **Causes**:
  - Inadvertent contact with teeth during physical altercation
  - Human bite
- Common Pathogens:
  - *Streptococcus viridians*
  - *Streptococcus pyogenes (Group A Strep)*
  - *Staphylococcus aureus*
  - *Eikenella corrodens*

Fight bite injury

When the patient with a fight bite injury is examined with his or her fingers extended, the small skin laceration may no longer correspond to the fracture site and is frequently overlooked because of the innocuous clinical appearance (arrow).

UpToDate®

Ganglion Cyst

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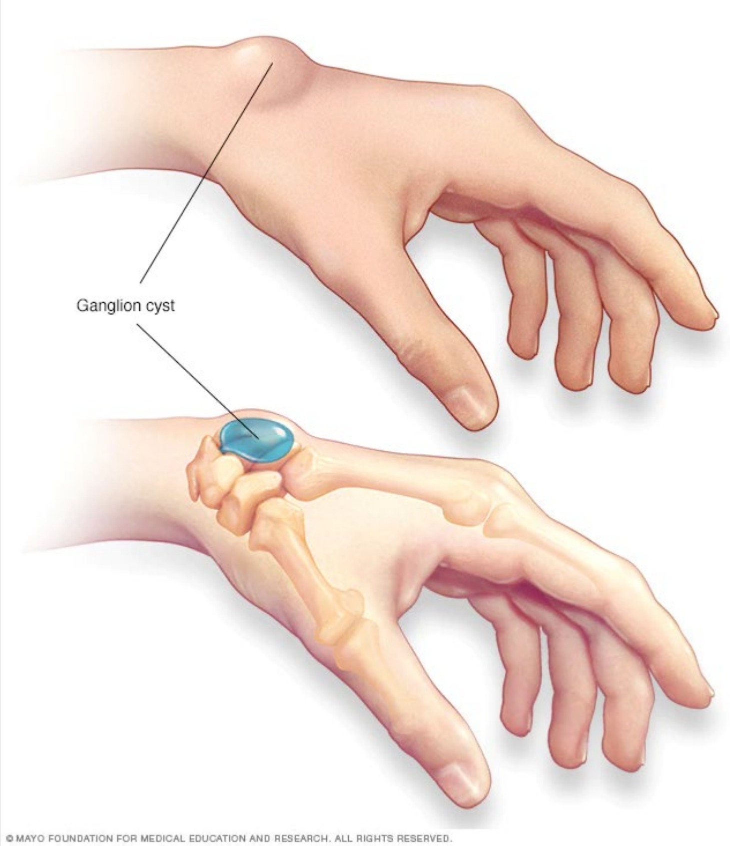
- Fluid filled swelling overlying the joint or tendon sheath
  - Fluid is typically from joint or tendon sheath
- Mechanism: trauma or synovial herniation
- Most common location is at the dorsal wrist (70%)
- **Typically painless, but a palpable bump that transilluminates on physical exam**
  - This feature is key in differentiating from other causes such as solid tumors

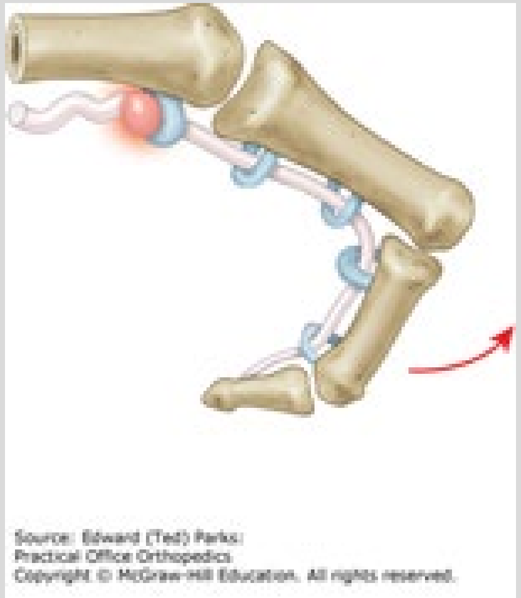
- No indication for removal, as most spontaneously resolve
  - If remains, cyst may require aspiration or surgery for removal if patient presents with bothersome symptoms

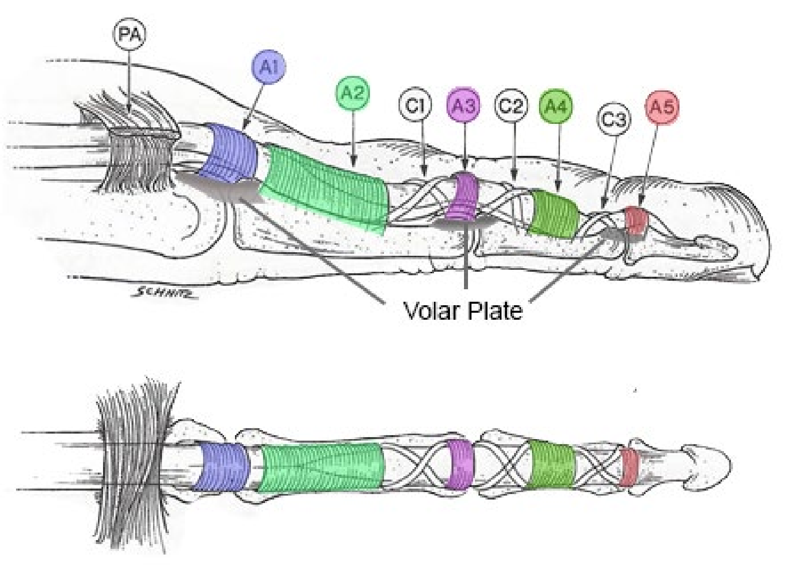


The Hand

Trigger Finger

- **Presentation:** Presents as the catching and locking of the finger when flexed
  - Caused by stenosing tenosynovitis of the flexor tendons
  - Pain and catching of the finger when flexed, with the finger getting locked in flexion
- **Mechanism**: mechanical impingement at the level of A1 pulley of the hand
  - Normal gliding of the tendons are disrupted, leading to increased resistance as the tendon pulls through the A1 pully, leading to tendon entrapment
  - Once flexed, the affected finger will remain locked due to the mechanical impingement
- **Risk Factors:** diabetes mellitus





Mallet Finger

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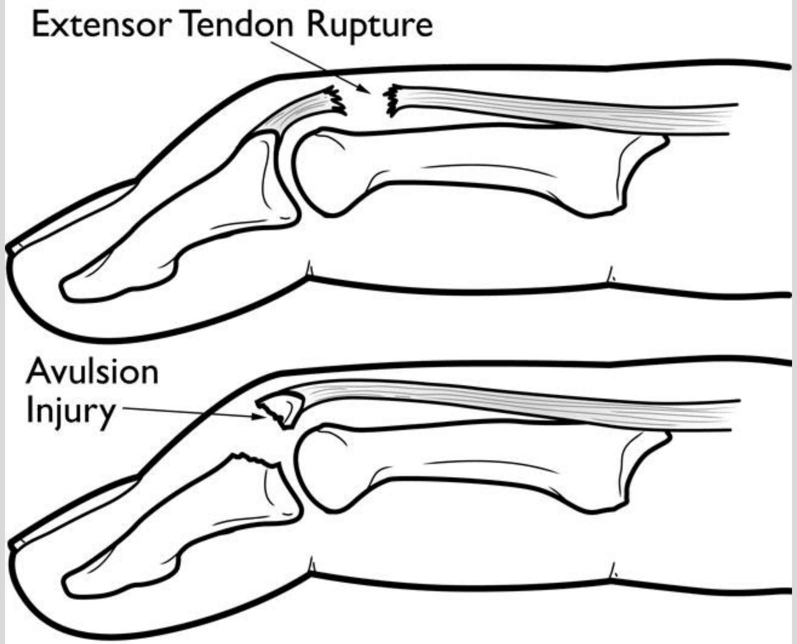
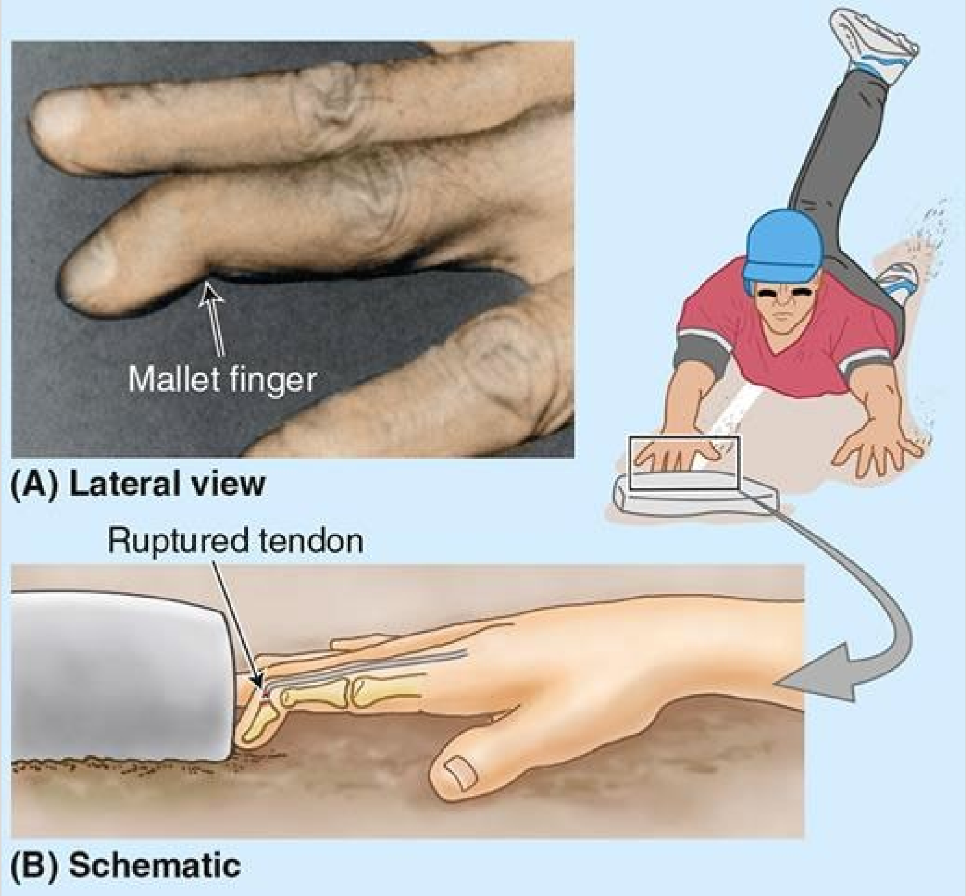
- **Presentation:** prolonged flexion of the distal interphalangeal joint (DIP)
- **Mechanism**: finger deformity caused by disruption of the terminal extensor tendon distal to the DIP joint
  - A blow to the distal tip of the digit causes rupture of the extensor tendon
  - Also caused by avulsion fracture
- **Treatment: splinting**



https://orthoinfo.aaos.org/en/diseases--conditions/mallet-finger-baseball-finger/

Moore's Clinically Oriented Anatomy, 9e



Carpal Tunnel

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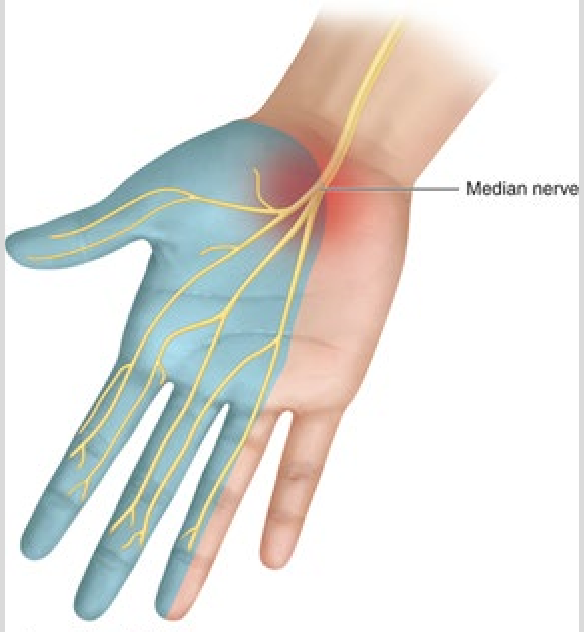
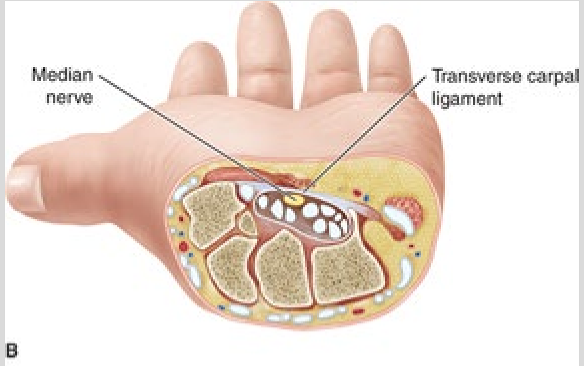
- **Mechanism**: entrapment of the **median nerve** in the carpal tunnel
  - Carpal Tunnel consists of foramen between **transverse carpal ligament** and carpal bones
  - Can be due to lunate dislocation
- **Presentation**: paresthesia, pain, numbness in distribution of median nerve
  - Thenar eminence atrophy but sensation remains in tact
  - Muscles controlled by median nerve, but palmar cutaneous branch enters the hand external to the carpal tunnel
- **Physical Exam:**
  - **Tinel Sign:** percussion of wrist causes tingling
  - **Phalen Maneuver**: flexion of wrists cause tingling
- **Associations:**
  - Pregnancy (edema)
  - Rheumatoid arthritis, hypothyroidism, diabetes, acromegaly, amyloidosis
  - Repetitive use!



Source: Edward (Ted) Parks:
Practical Office Orthopedics



Source: Edward (Ted) Parks:
Practical Office Orthopedics

Carpal Tunnel Tests

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Phalen Test

Tinel Sign

Thenar Atrophy

Source: Edward (Ted) Parks:
Practical Office Orthopedics

Guyon Canal Syndrome

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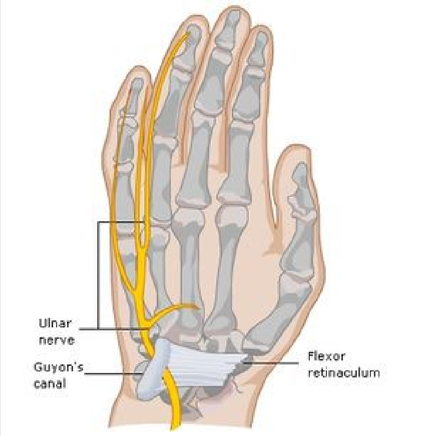
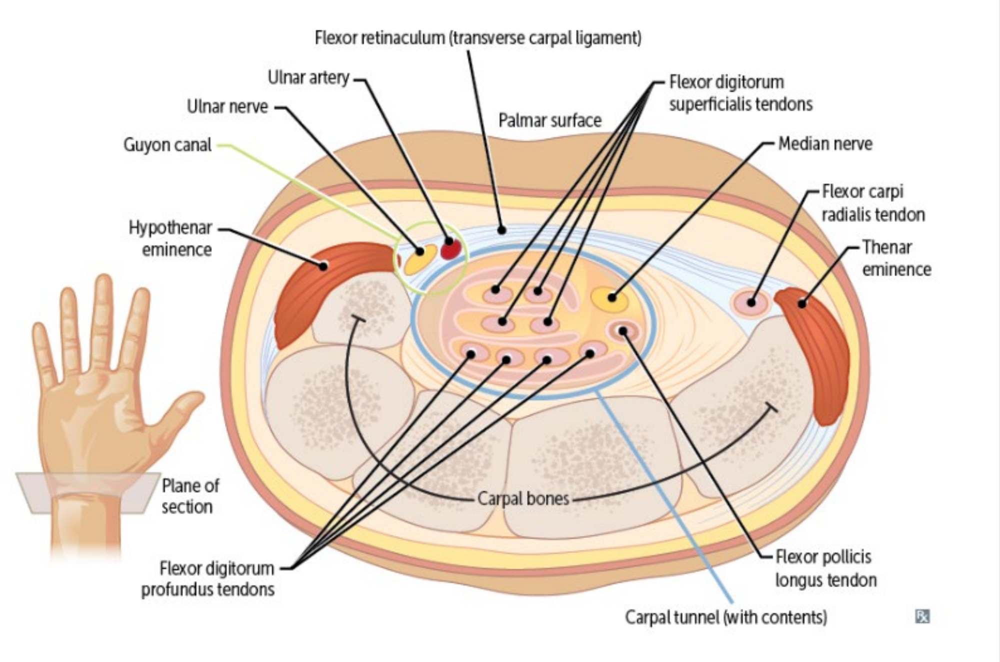
- **Mechanism**: compression of the **ulnar** nerve at the wrist
  - Can be due to hook of hamate fracture
- **Presentation**:
  - Classically seen in **cyclists** due to pressure at the wrist
  - Motor symptoms due to **muscles effected by ulnar nerve** innervation
    - **Intrinsic muscles of the hand** (decreased handgrip and clawing of the fourth/fifth digits)
    - **Hypothenar atrophy**
- Comparison to cubital tunnel syndrome:
  - **Guyon Canal Syndrome will have NORMAL** function of the extrinsic (forearm) muscles supplied by the ulnar nerve
    - Ulnar n innervates flexor carpi ulnaris and flexor digitorum profundus





De Quervian’s Tenosynovitis

All Peanut Lovers Enjoy Peanut Butter

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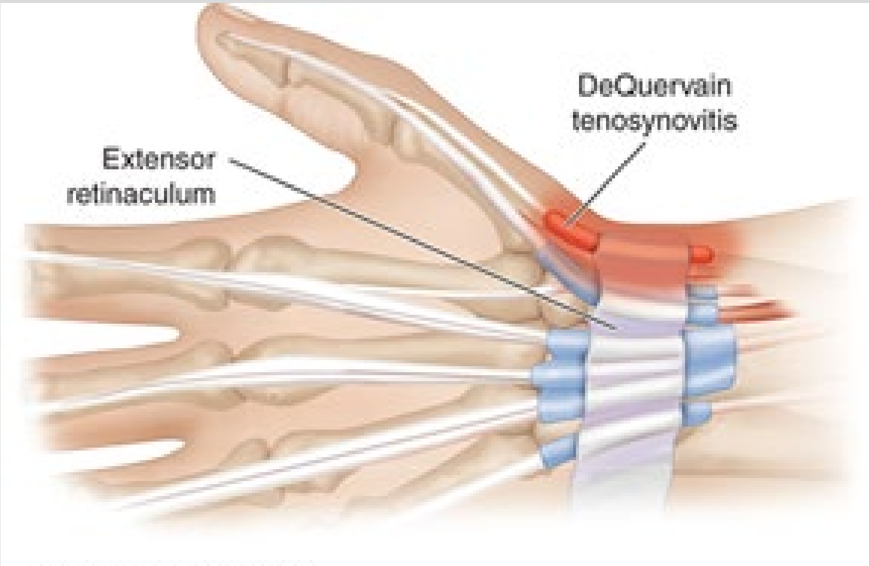
- **Mechanism**: thickening of the **abductor pollicis longus and extensor pollicis brevis tendons (APL + EPB)**
  - Due to excessive friction, trauma, or inflammation
  - APL and EPB tendons will rub on extensor retinaculum creating this irritation in the tendons
  - Common in **NEW PARENTS** as result of increased friction from holding their child
  - Also seen in racquet sports and golfers
- **Presentation**:
  - Pain at the radial styloid
  - Swelling
- **Physical Exam:**
  - Finkelstein’s test: patient places their thumb in their palm, wrapping their fingers around their thumb to make a fist with their thumb tucked inside. Patient will ulnar deviate the hand towards their pinky finger. Positive sign is pain
- **Treatment**:
  - Spica split

*PowerPoint edited from recorded version. It should read “EPB,” not “EBP.” MH



Source: Edward (Ted) Parks: Practical Office Orthopedics

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Tendon Sheath Infection: Septic Flexor Tenosynovitis

- **Pathophysiology:** infection of the synovial sheath that surrounds the flexor tendon
  - Typically due to a penetrating trauma of the tendon sheath
- **Presentation:** Kanavel Signs
  - **Flexed posturing of the digit**
  - **Tenderness to palpation over the tendon sheath**
  - **Marked pain with extension of digit**
  - **Fusiform swelling of joint**
- **Treatment:** antibiotics and surgery to clean and irrigate the infected tendon sheath

Orthobullets

Practice Questions

Question #1

NEW YORK INSTITUTE OF TECHNOLOGY

A 50 year old man reports pain near the thumb side of the wrist. There is also visible swelling near the base of the thumb. The patient mentions it is hard to pick objects up and grasp them with the help of his thumb. The doctor suspects a thickening of the abductor pollicis longus and extensor pollicis brevis tendons. Which of the following would be a good test to confirm this suspicion?

A. Neer's Test
B. Empty can Test
C. Hawkins Test
D. Finkelstein's Test
E. Adson's Test

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Question #2

NEW YORK INSTITUTE OF TECHNOLOGY

A 6 year old girl arrives to the local urgent center after falling in the park. She is holding her arm in pain and pointing near the distal humerus when asked where it is most painful. X-ray is taken and the results are shown below. Which of the following arteries are most likely to be damaged based on patient's injury?

A. Median
B. Ulnar
C. Brachial
D. Deep brachial
E. Radial

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Question #3

NEW YORK INSTITUTE OF TECHNOLOGY

A 45-year-old man presents to clinic with a two month history of shoulder pain that is worse with overhead activity, and especially bothersome at night. On exam, he complains of pain to palpation just below the acromion. An empty can test is positive, and you suspect he has torn his supraspinatus. If correct, which of these functional maneuvers would you expect to be deficient on physical exam?

A) External rotation
B) Initiation of abduction
C) Completion of abduction
D) Internal rotation
E) Circumduction

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Question #4

NEW YORK INSTITUTE OF TECHNOLOGY

A 40 year old male presents the emergency department after falling off the ladder while fixing the roof of his home. He reports to falling with his weight on top of his arm. X-ray is shown below. What deficits can be observed on physical exam?

A. Inability to flex the wrist
B. Inability to extend the wrist
C. Loss of sensation to the 4^{th} and 5^{th} digit
D. Loss of sensation over the thenar eminence
E. Inability to abduct the arm

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Question #5

NEW YORK INSTITUTE OF TECHNOLOGY

A 48 year old female accountant presents to an orthopedist regarding her wrist pain. She states that when towards the end of the day, she feels a shooting nerve pain followed by numbness. The pain is particularly worse at night Upon physical examination, the doctor notices that she has noticeable thenar atrophy but sensation is spared. The nerve that is implicated in this disease process runs under what structure?

A. Guyon's Canal
B. Flexor Retinaculum
C. A1 pulley
D. Brachial Fascia
E. Axillary artery

Question #6

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Question #7

NEW YORK INSTITUTE OF TECHNOLOGY

A 30-year-old female presents to the clinic complaining of wrist and hand pain. She states the pain started two weeks ago, and is 36-weeks pregnant. Percussion over the volar aspect of the wrist elicits tingling in the first three digits. On neurological exam, sensation is spared over the thenar muscles. Which nerve results in spared sensation in this given condition?

A. Ulnar nerve
B. Radial Nerve
C. Palmar cutaneous branches of Median nerve
D. Musculocutaneous Nerve
E. Anterior Interosseous Nerve

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Question #8

NEW YORK INSTITUTE OF TECHNOLOGY

A 3-year-old male has decreased use of his left elbow after his mother grabbed his arm and attempted to lead him across the street. Physical exam demonstrates guarding of the extremity with a slightly flexed and pronated arm, elbow swelling, and focal tenderness. What is the most likely reason for this presentation?

A. Fractured Scaphoid
B. Clavicular dislocation
C. Clavicular Fracture
D. Midshaft humeral fracture
E. Subluxed radial head

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Summary Slide

NEW YORK INSTITUTE OF TECHNOLOGY

- General topics
  - Compartment syndrome: 6 Ps, fasciotomy
  - Fracture basics, pediatric fractures
- Shoulder
  - Clavicle fracture vs. spinal accessory nerve injury
  - Anterior vs posterior dislocations
  - Rotator cuff muscles, actions, innervations
- Humerus
  - Fracture locations and which neurovascular involvement
- Elbow
  - Fractures, dislocations, bursitis
  - UCL injuries
  - Radial Head Subluxation - mechanism, presentation, treatment
- Forearm
  - Epicondylitis
  - Distal radius fractures
- Wrist
  - Anatomy, scaphoid fractures (mechanism, complications), lunate injury, boxer’s fracture, trigger finger
- Hand
  - Mallet finger, Carpal Tunnel Syndrome, Guyon’s Canal Syndrome, De Quervain’s tenosynovitis

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Lecture Feedback:

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